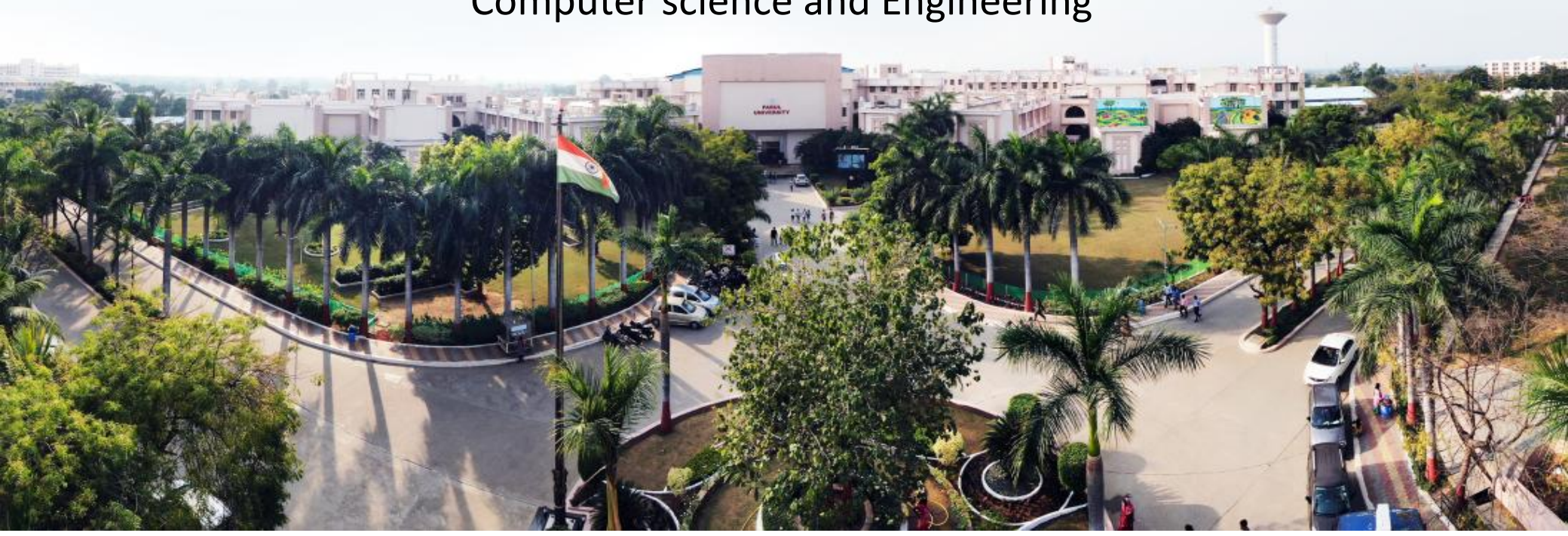


Image Processing (303105381)

Mr. Anurag Kewat

Lecturer
Computer science and Engineering



UNIT-1

Digital Image Fundamentals

Outline

1. Objectives and applications of image processing
2. Fundamental steps in digital image processing
3. Components of Image Processing System
4. Elements of Visual perception
5. Light and the Electromagnetic Spectrum
6. Image sensing and acquisition
7. Image sampling and quantization
8. Basic relationships between Pixels.
9. Linear and Nonlinear Operations.

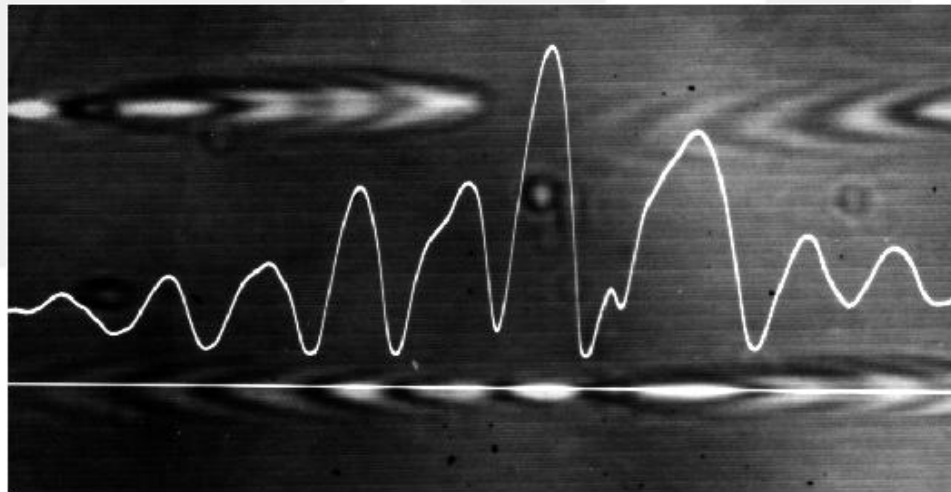
Outcome of this unit:

After completing the students will be able to:

- To understand the fundamental concepts of Digital Image Processing.
- To learn basic image processing operations.
- To understand image analysis algorithms.
- To get exposure of current applications in the field of image processing.

1. What is Digital Image Processing

- The field of *digital image processing* refers to **processing digital images by means of a digital computer**.
- A digital image can be defined as a two-dimensional function, $f(x, y)$, where x and y are *spatial* coordinates, and f is a *intensity* or *gray level* of the image at that point.



Cont...

- Definition and importance of digital image processing

Digital Image Processing

- Digital image is composed of a finite number of elements, each of which has a particular location and value.
- An image is denoted by a two dimensional function of the form $f(x, y)$
- We call the intensity of an image at any coordinates (x, y) the gray level (I) of the image at that point $I = f(x, y)$
- These elements are called picture elements, image elements, pels and pixels.
- Pixel is the term used most widely to denote the elements of digital image. A **pixel** has a location (the spatial coordinate) and a value.

Contd...

Image Processing does:

- Manipulation of digital images by computer.
- A preprocessing step necessary for image analysis and understanding.
- Digital images can be from many types of sources.
- Digital image processing is closely related to **signal processing** at one end, and to **computer vision** at the other end.
- Related areas
 - Computer Vision
 - Computer Graphics
- Applications in various fields such as medical imaging, satellite imaging, security, etc.

Motivation

Image processing is mainly used for two main purposes:

- Improving the visual appearance of images (pictorial information) to a human viewer, and
- Preparing (processing) images for measurement of the features and structures present from autonomous machine perception.
- This course covers methods that are used for both purposes.

Computer vision

- **Computer vision** is concerned with making computers “**understand**” what is contained in an image and other functions of human vision.
- It is sometimes considered to be a topic of **artificial intelligence**.
- However, it is not possible to state exactly where image processing ends and computer vision begins.

❖ Image Processing

image in -> image out

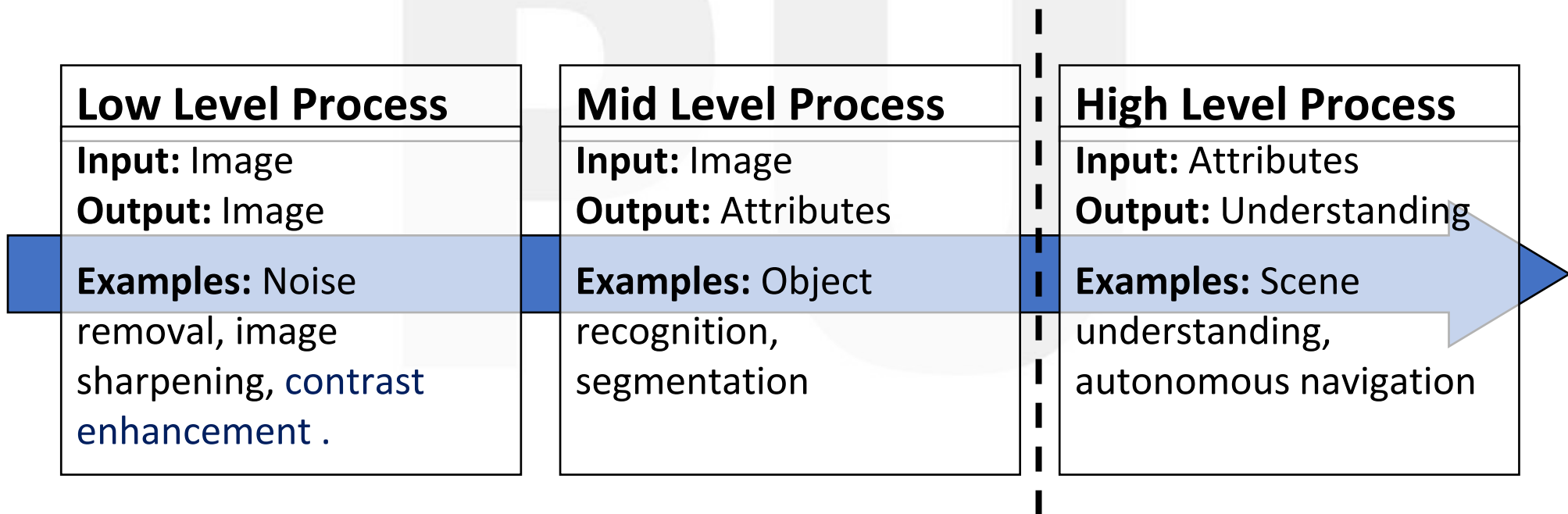
❖ Image Analysis

image in -> measurements out

❖ Image Understanding

image in -> high-level description out

There are three types of computerized processes in the processing of image low-, mid- and high-level processes



Applications of Digital Image Processing (DIP)

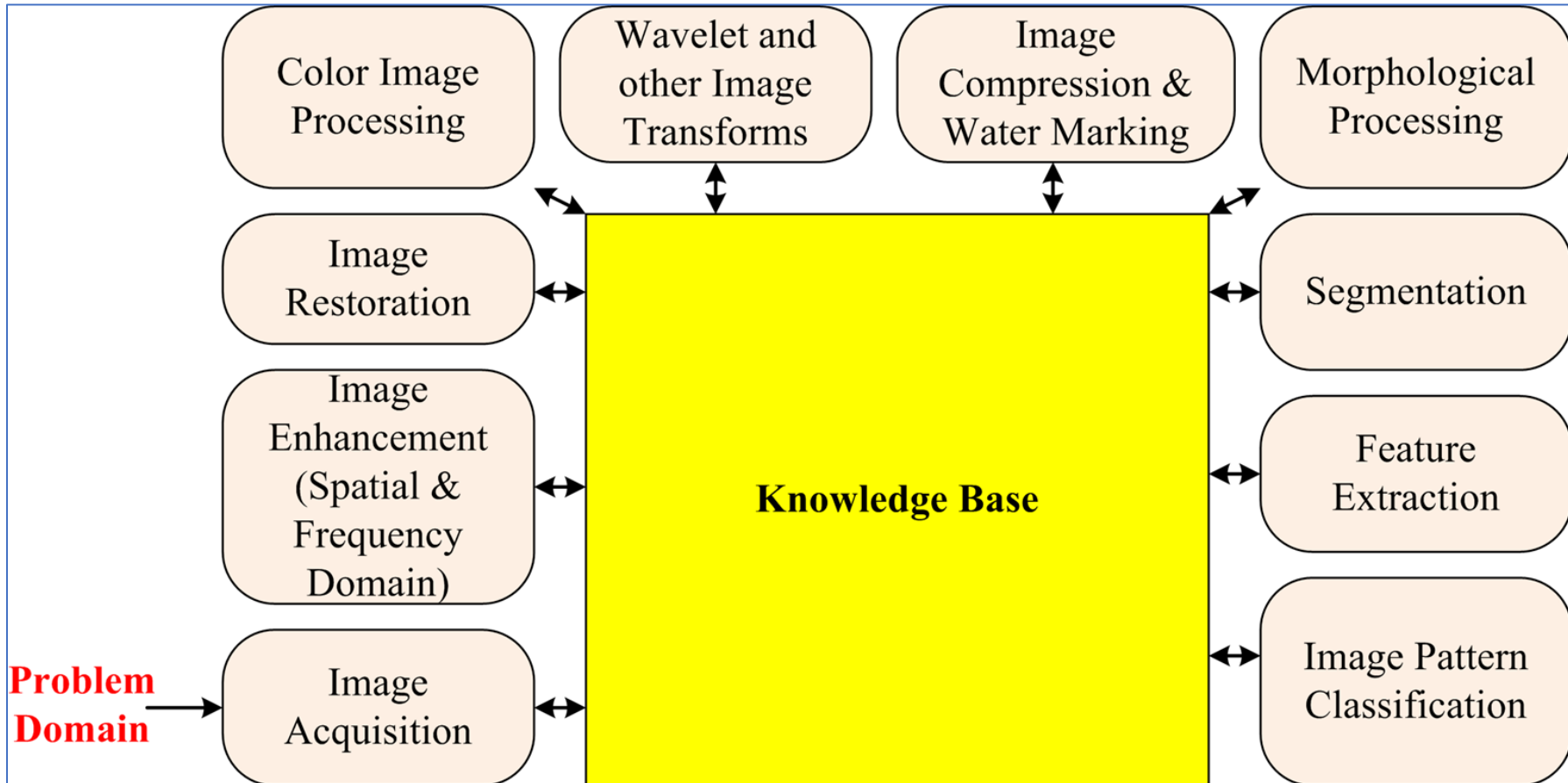
Modern applications of this technology range from simple tasks such as resizing photos to complex applications including medical imaging diagnostics, autonomous vehicles' s visual systems, and facial recognition in security systems:

- **Biological Research:** e.g. DNA typing and matching; automatic counting and classification of cell structures in bone and tissue.
- **Defense and Intelligence:** e.g. Reconnaissance photo-interpretation of objects in satellite images; target acquisition and missile guidance.
- **Document Processing:** e.g. Scanning, archiving and transmission (fax); automatic detection and recognition of printed text (postal sorting office, tax return processing, banking cheques).

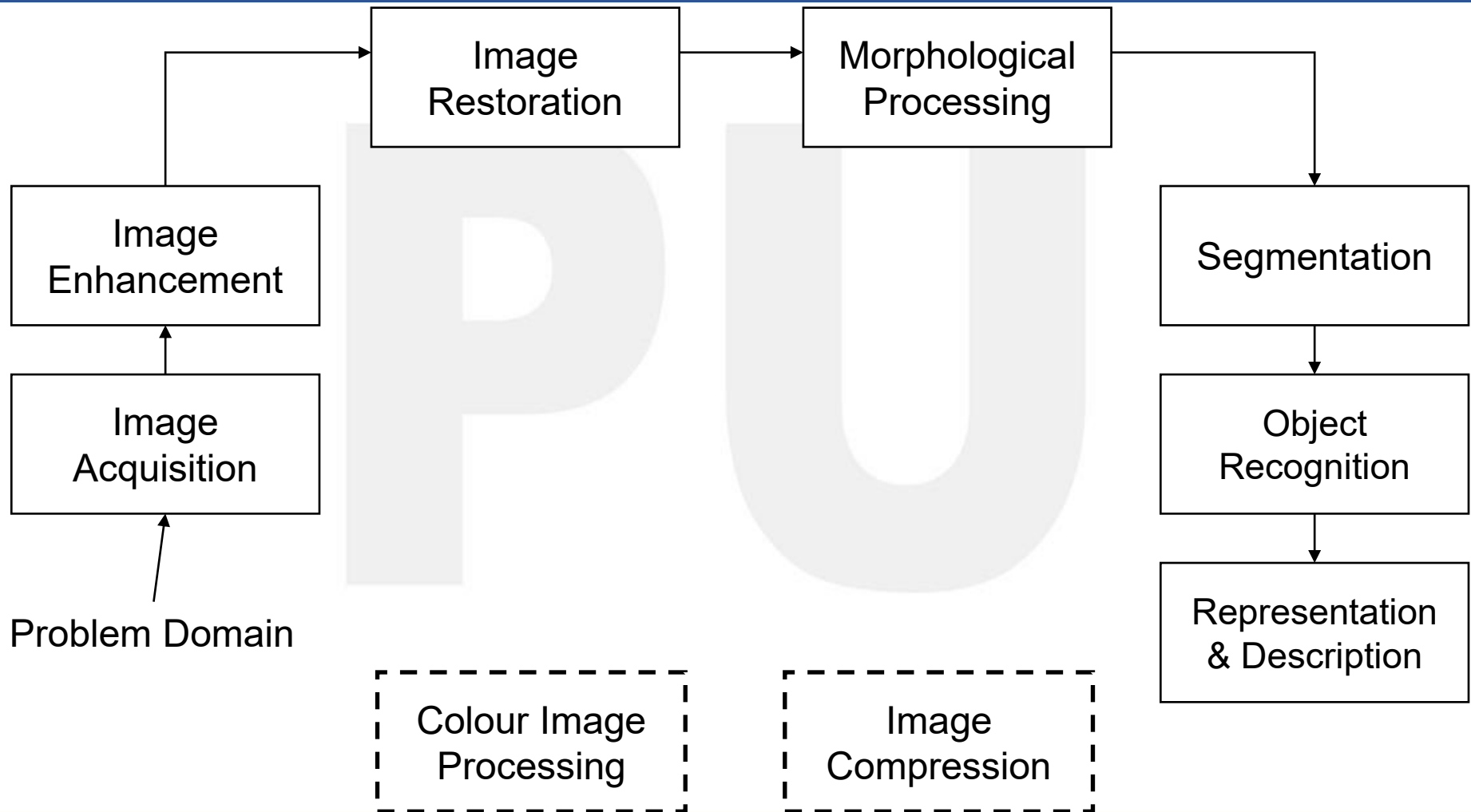
- **Factory Automation:** e.g. Visual quality inspection, defect detection and process monitoring.
- **Law Enforcement Forensics:** e.g. Photo-ID kits, criminal photo-search, automatic fingerprint matching, DNA matching and fiber analysis.
- **Materials Research:** e.g. Automatic counting and classification of object characteristics such as **impurities and grain size**; surface and structural defect analysis (**x-ray analysis for aircraft wing cracks**)
- **Photography:** e.g. Retouching defects, altering colors, zooming; adding and subtracting objects to a scene; special effects such as blending and warping.
- **Publishing:** e.g. Layout composition, **inserting pictures, generating graphics; colour separation for 4-colour printing** (cyan, magenta, yellow and black).

- **Remote Sensing:** e.g. Land cover analysis (water, roads, cities and cultivation), vegetation features (water content and temperature) and crop yield analysis; 3-D terrain rendering from satellite or aircraft data (road and dam planning); fire and smoke detection.
- **Space exploration and Astronomy:** e.g. Image compression for transmission, correction of detector deficiency; automatic satellite navigation and altitude control using star positions.
- **Video and Film Special Effects:** Animation, dangerous stunts (explosions) and special effects (Star Wars).
- **Other examples/areas ??**

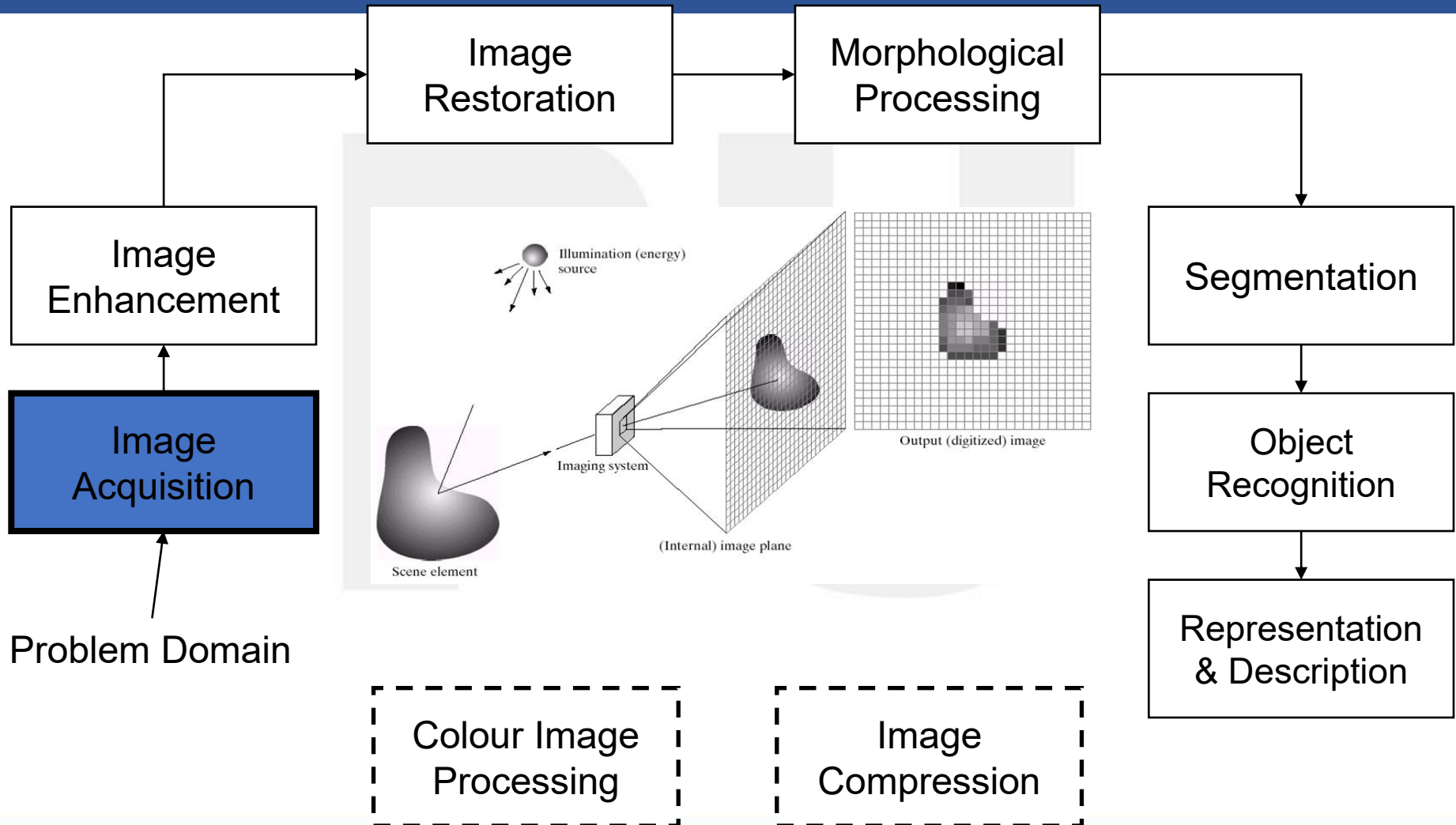
2. Fundamental Steps in Digital Image Processing



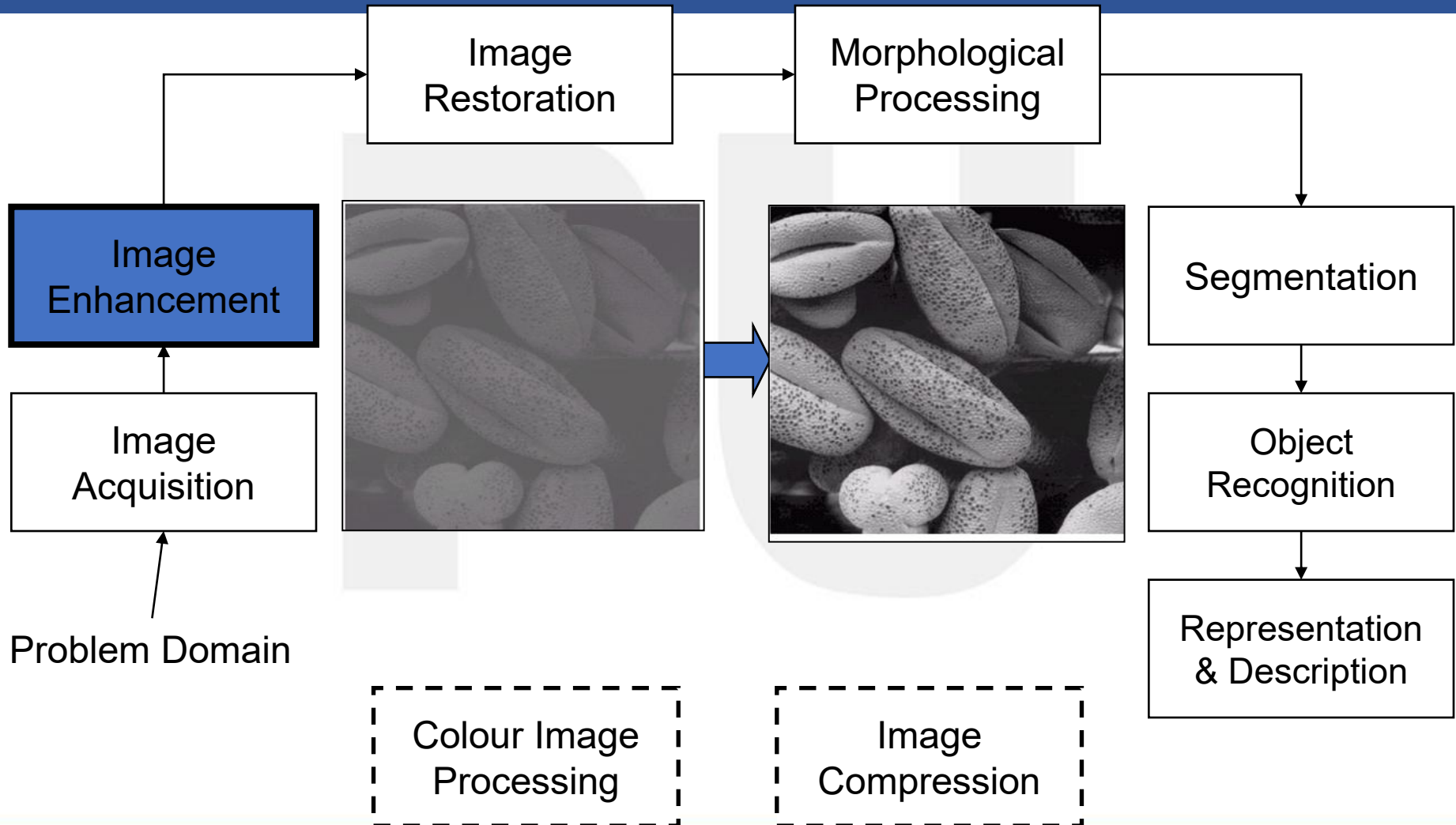
Key Stages in Digital Image Processing



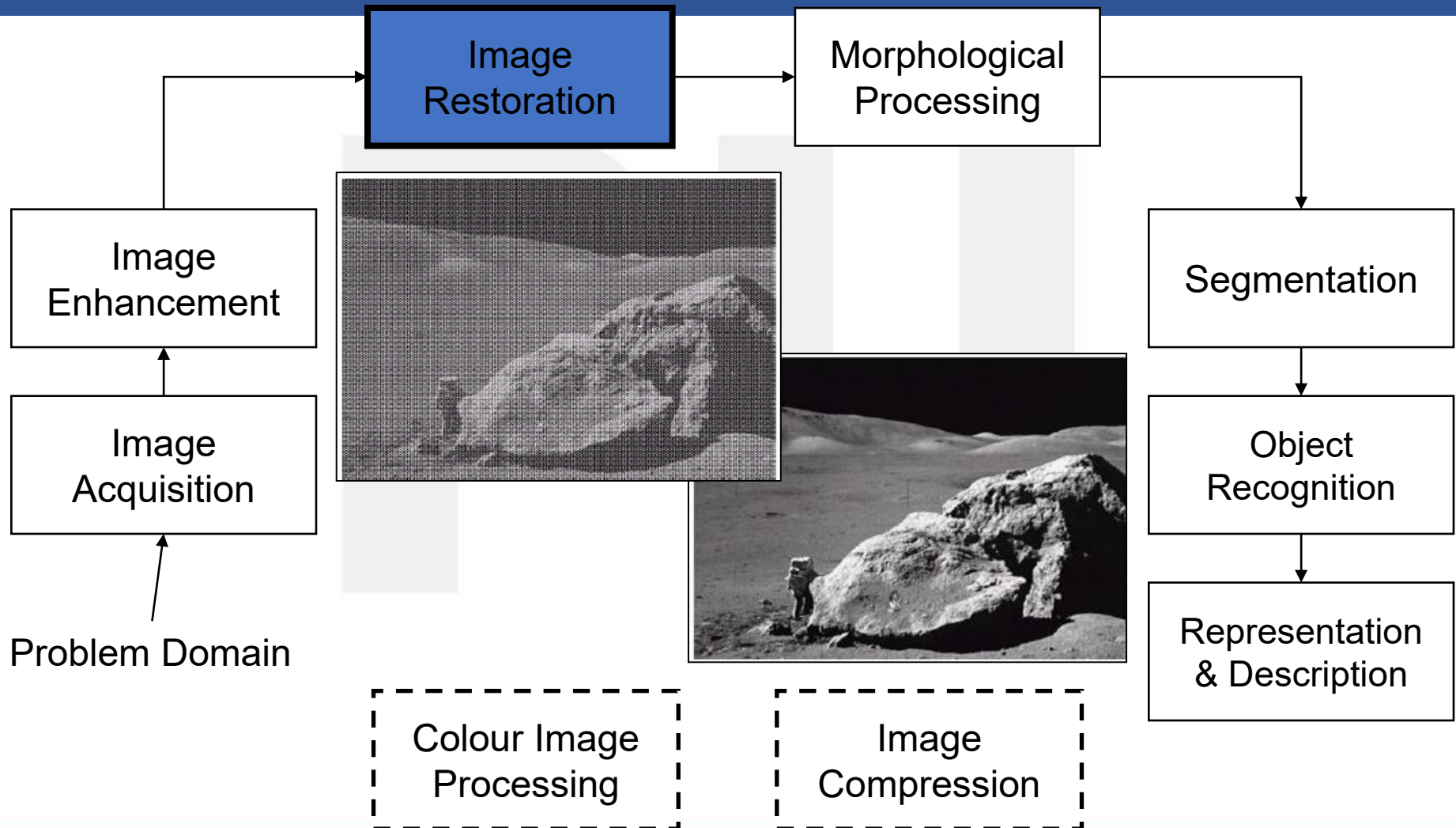
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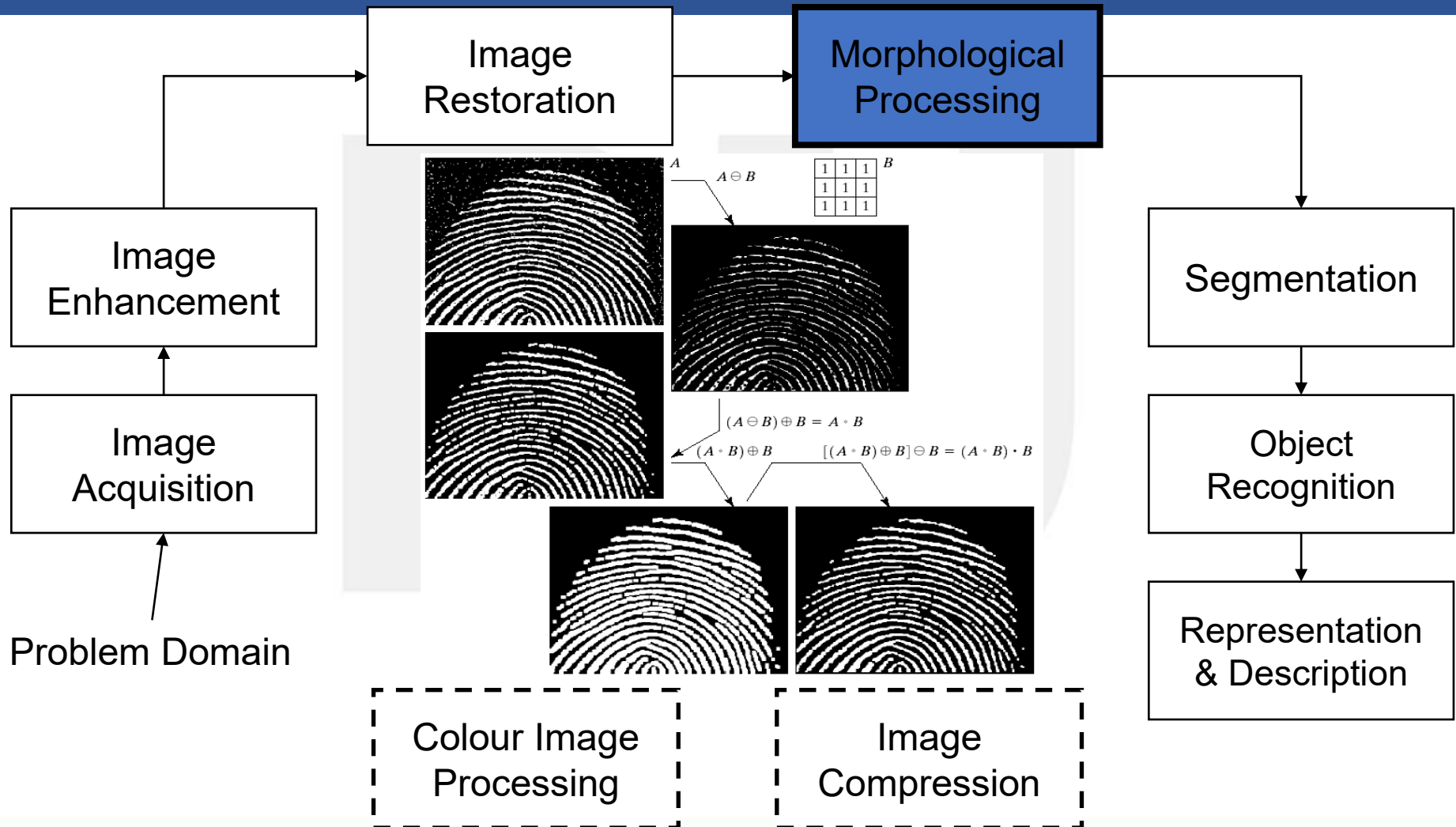
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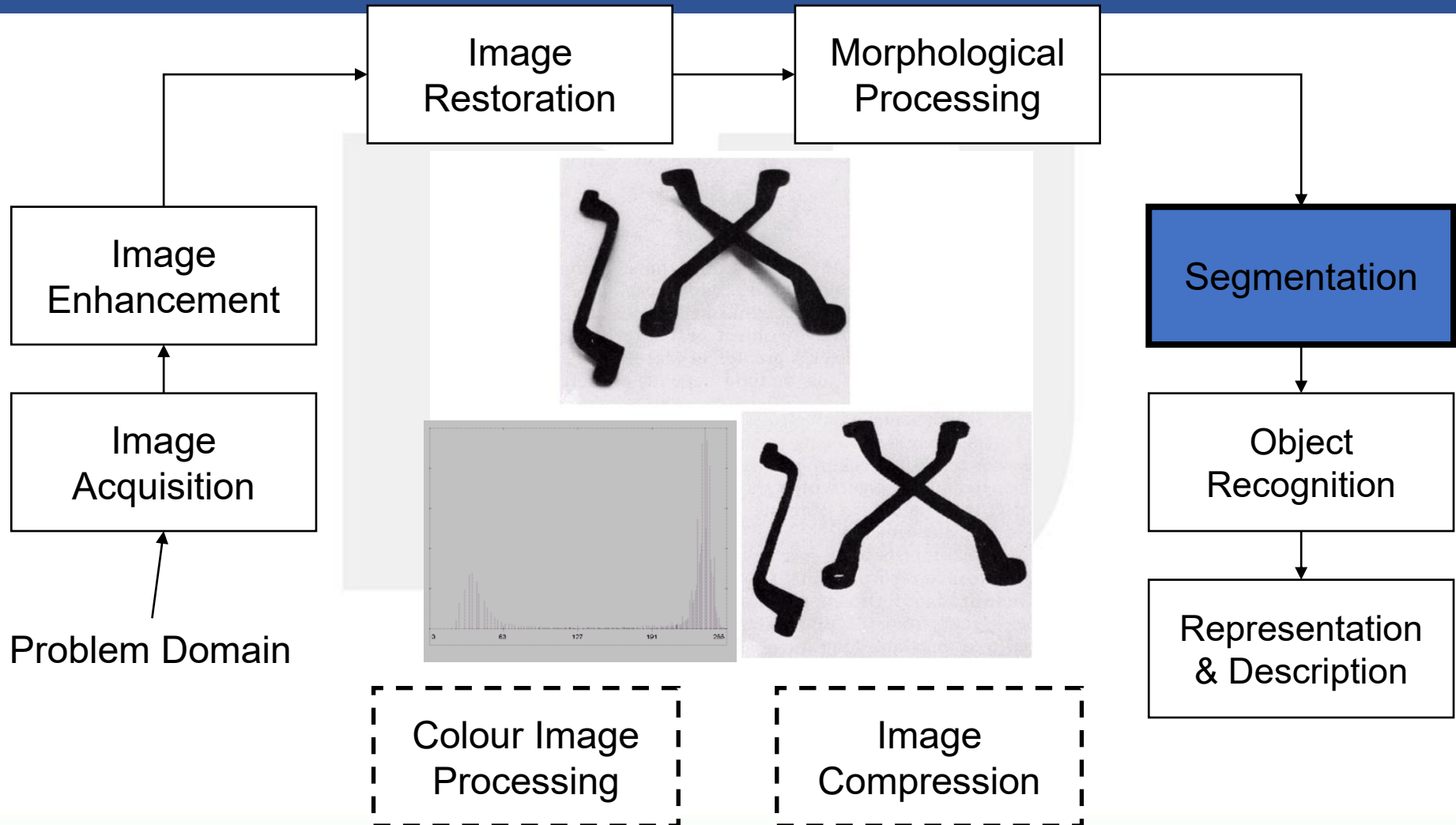
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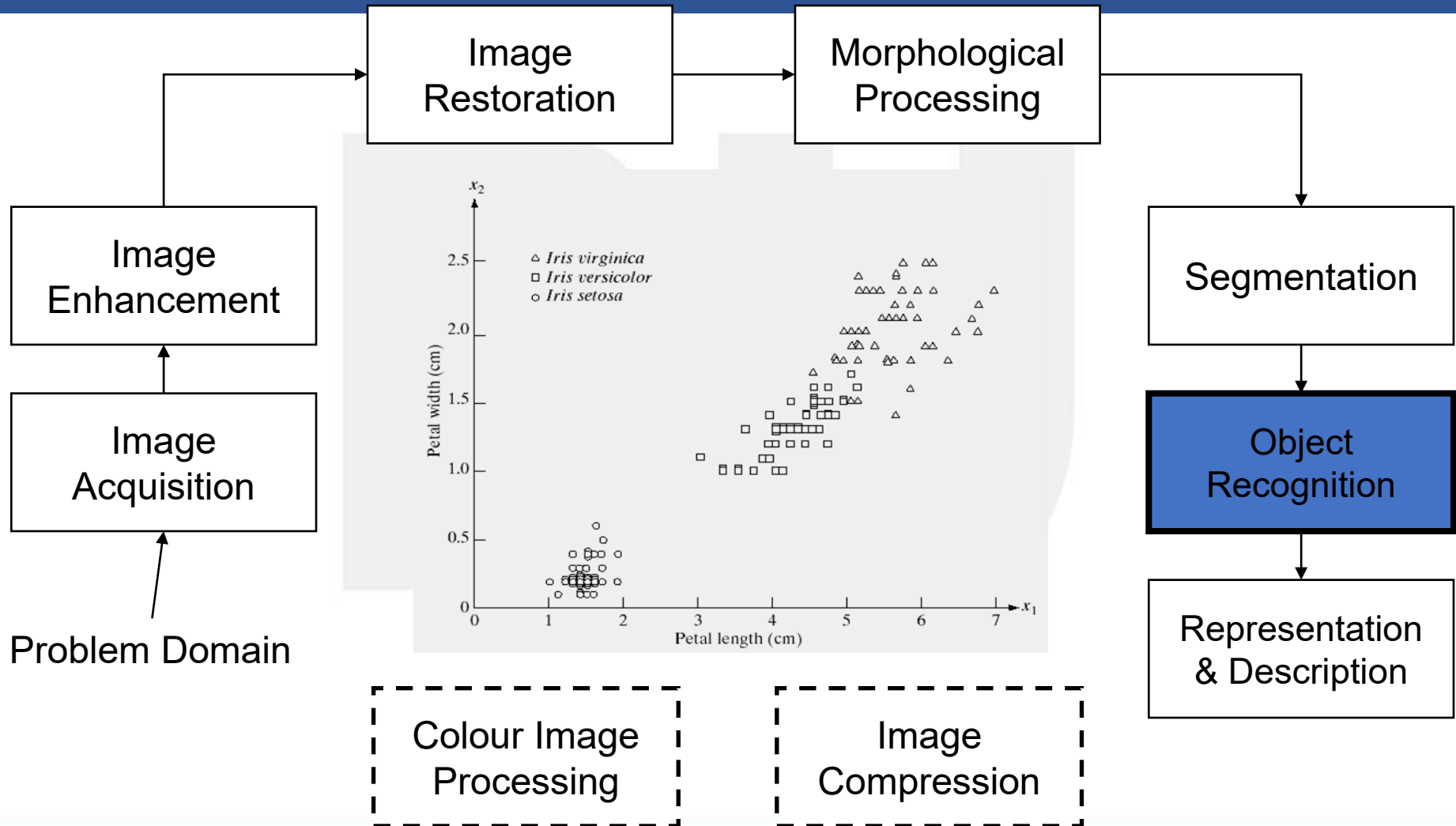
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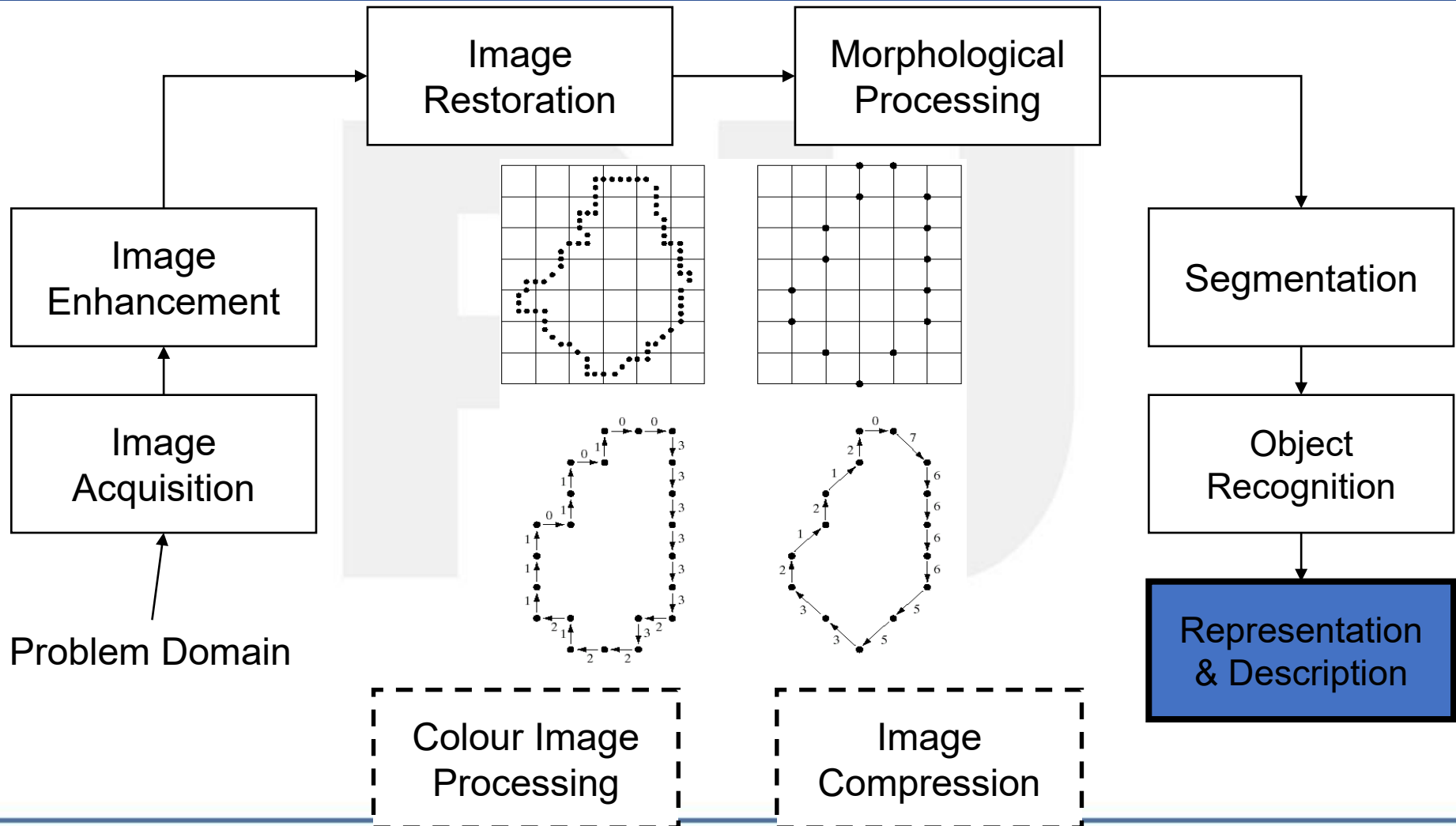
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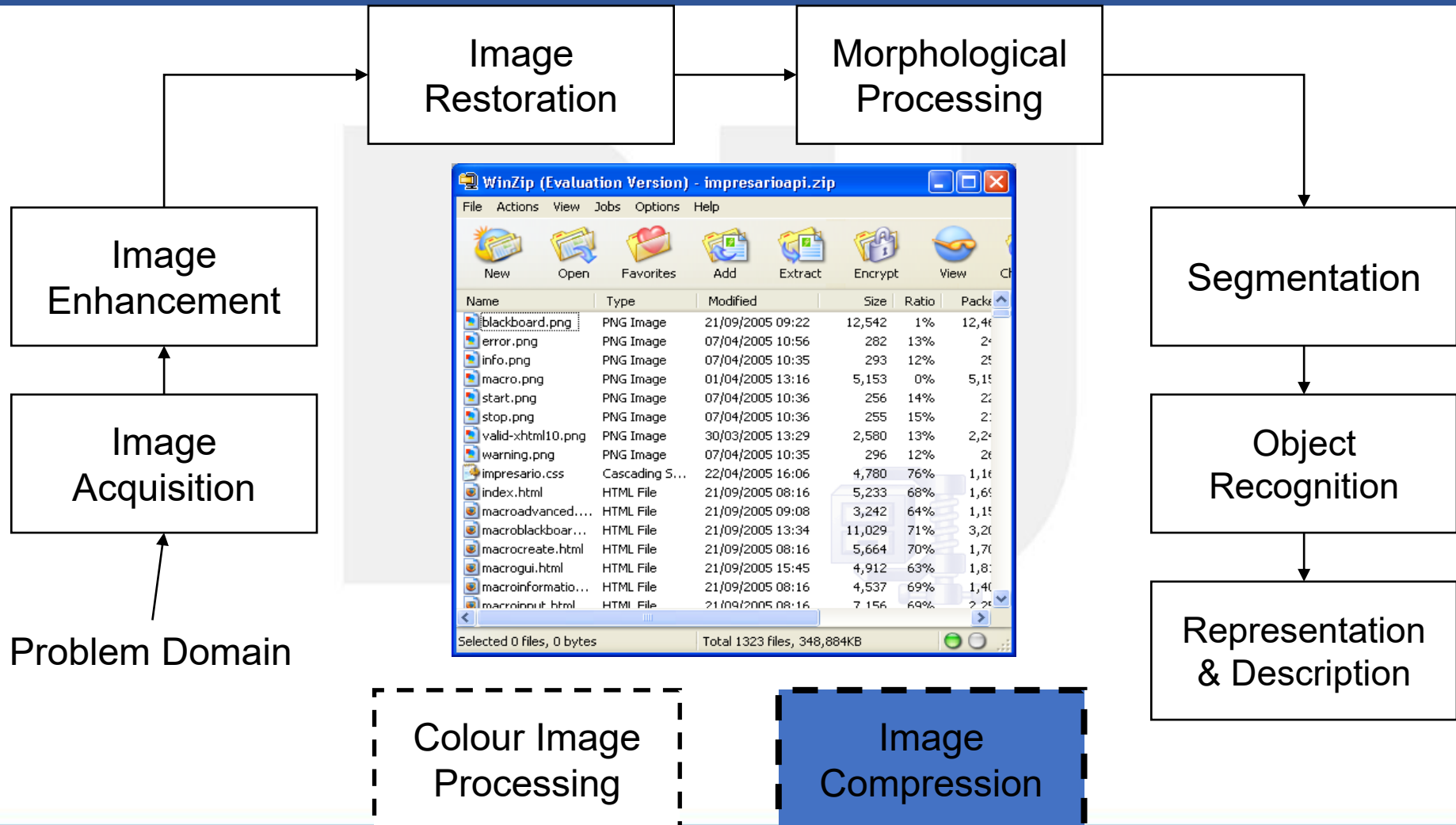
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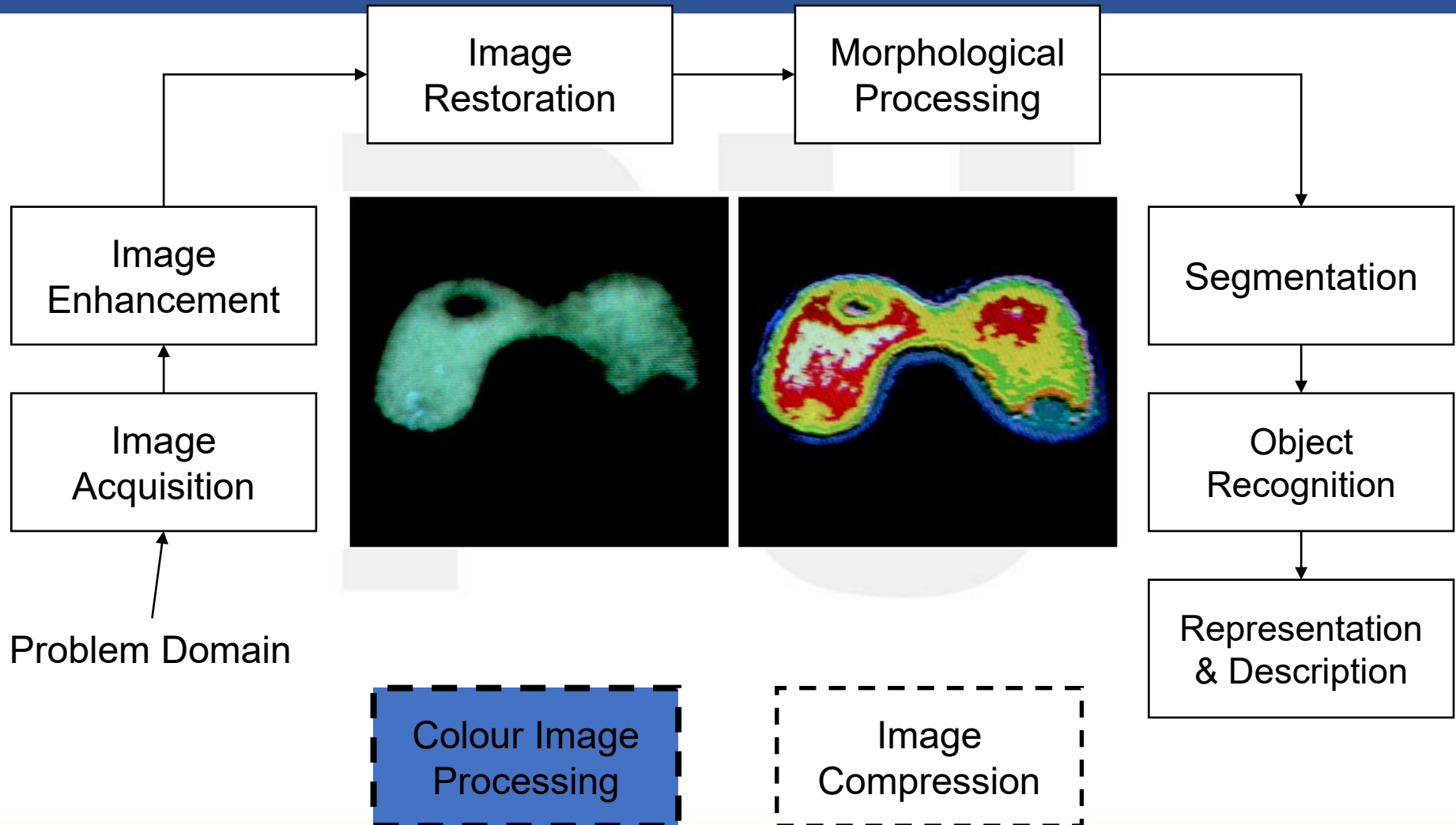
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3. Components of an Image Processing System

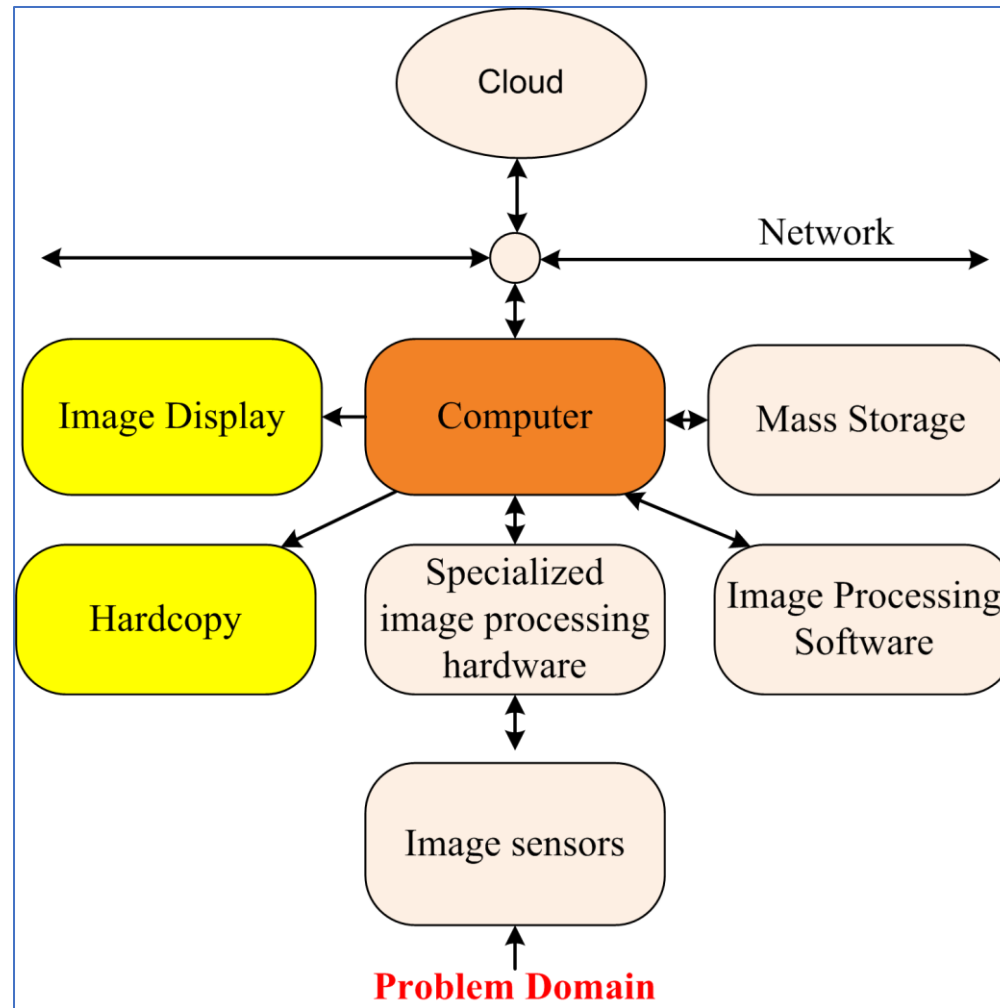


Image Sensor

For sensing, two elements are required to acquire digital images.

1. A physical device that is sensitive to the energy radiated by the object we wish to image.
2. A digitizer, is a device for converting the output of the physical sensing device into digital form.

For instance, in a digital video camera, the sensors produce an electrical output proportional to light intensity. The digitizer converts these outputs to digital data.

Specialized image processing hardware

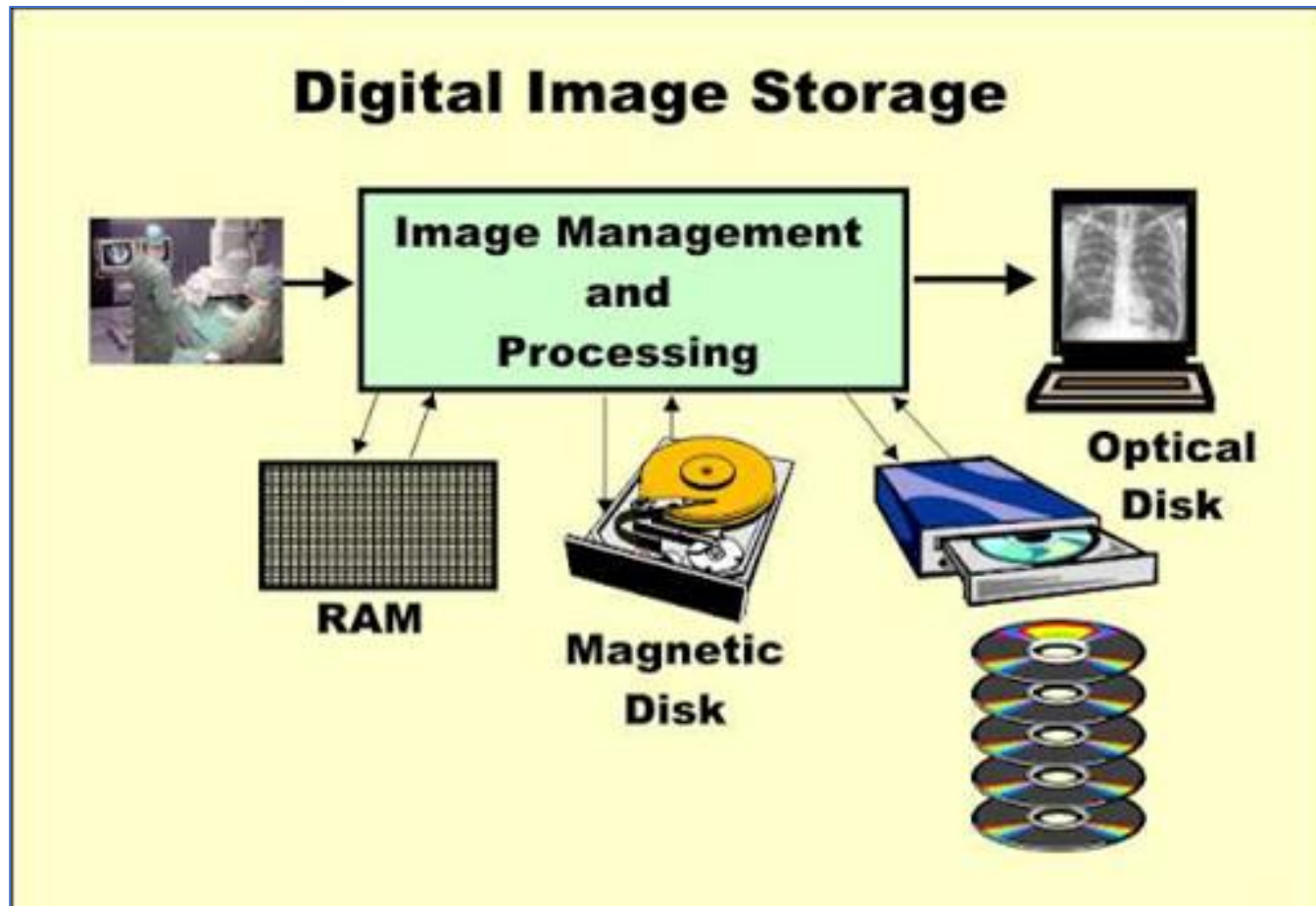
- It usually consists of the digitizer, plus hardware that performs other primitive operations, such as an arithmetic logic unit (ALU), which performs arithmetic and logical operations in parallel on entire images.
- Example: An ALU is used in averaging images as quickly as they are digitized, for the purpose of noise reduction.
- This hardware is also known as a front-end subsystem, and its most distinguishing characteristic is speed. i.e., this unit performs functions that require fast data throughputs (e.g., digitizing and averaging video images at 30 frames/s) that the typical main computer cannot handle.

Software

- It consists of specialized modules that perform specific tasks.
- A well-designed package also includes the capability for the user to write code that, as a minimum, utilizes the specialized modules.
- More sophisticated software packages allow the integration of those modules and general-purpose software commands from at least one computer language.

Mass Storage

- When dealing with thousands, or even millions, of images, providing adequate storage in an image processing system can be a challenge.
- Digital storage for image processing applications falls into three principal categories:
 1. Short-term storage for use during processing,
 2. On-line storage for relatively fast re-call, and
 3. Archival storage, characterized by infrequent access.
- Storage is measured in bytes (eight bits), Kbytes (one thousand bytes), Mbytes (one million bytes), Gbytes (meaning giga, or one billion, bytes), and Tbytes (meaning tera, or one trillion, bytes).



<http://www.sprawls.org/resources/DIGSTORAGE/module.htm>

Image Display Device

- Image displays in use today are mainly color (preferably flat screen) TV monitors. Monitors are driven by the outputs of image and graphics display cards that are an integral part of the computer system.
- Seldom are there requirements for image display applications that cannot be met by display cards available commercially as part of the computer system.
- In some cases, it is necessary to have stereo displays, and these are implemented in the form of headgear containing two small displays embedded in goggles worn by the user.

Hardcopy

- Laser printers,
- Film cameras,
- Heat-sensitive devices, inkjet units, and
- Digital units, such as optical and CD-ROM disks.

The Film provides the highest possible resolution, but paper is the obvious medium of choice for written material.

- For presentations, images are displayed on film transparencies or in a digital medium if image projection equipment is used. The latter approach is gaining acceptance as the standard for image presentations.

Networking

- Networking is almost a default function in any computer system in use today.
- Because of the large amount of data inherent in image processing applications, the key consideration in image transmission is bandwidth.
- In dedicated networks, this typically is not a problem, but communications with remote sites via the Internet are not always as efficient.
- Fortunately, this situation is improving quickly as a result of optical fiber and other broadband technologies.

What is Digital Image Processing ?

Processing of a multidimensional pictures by a digital computer

Why we need Digital Image Processing ?

1. To record and store images
2. To enhance images using mathematics
3. To analysis images
4. To synthesize images
5. To create computer vision systems

Gray scale Image

When f , x and y are all finite and discrete quantities, the image is called a *digital image*

$$I = f(x, y)$$

Gray level
digital image

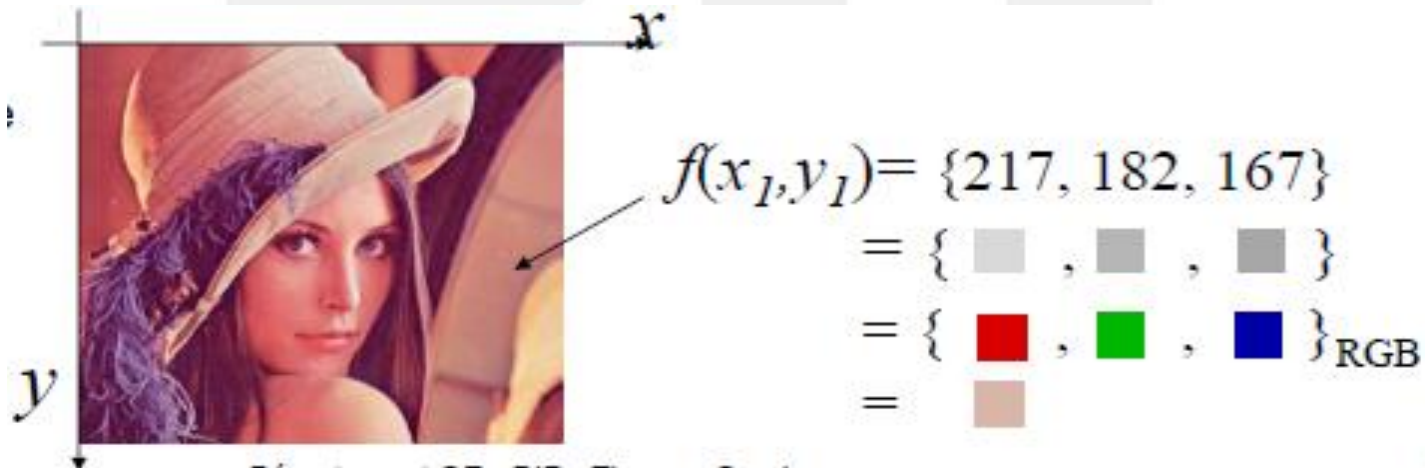


$$f(x_1, y_1) = 179 = \blacksquare$$

Color Image

The definition of f may be extended: as a n -dimensional function, i.e. 3D: $f(x,y,z)$ or image séquence $f(x,y,t)$ with amplitudes composed as a vector of data,

i.e. Color image: 3 components at each point, Complex number

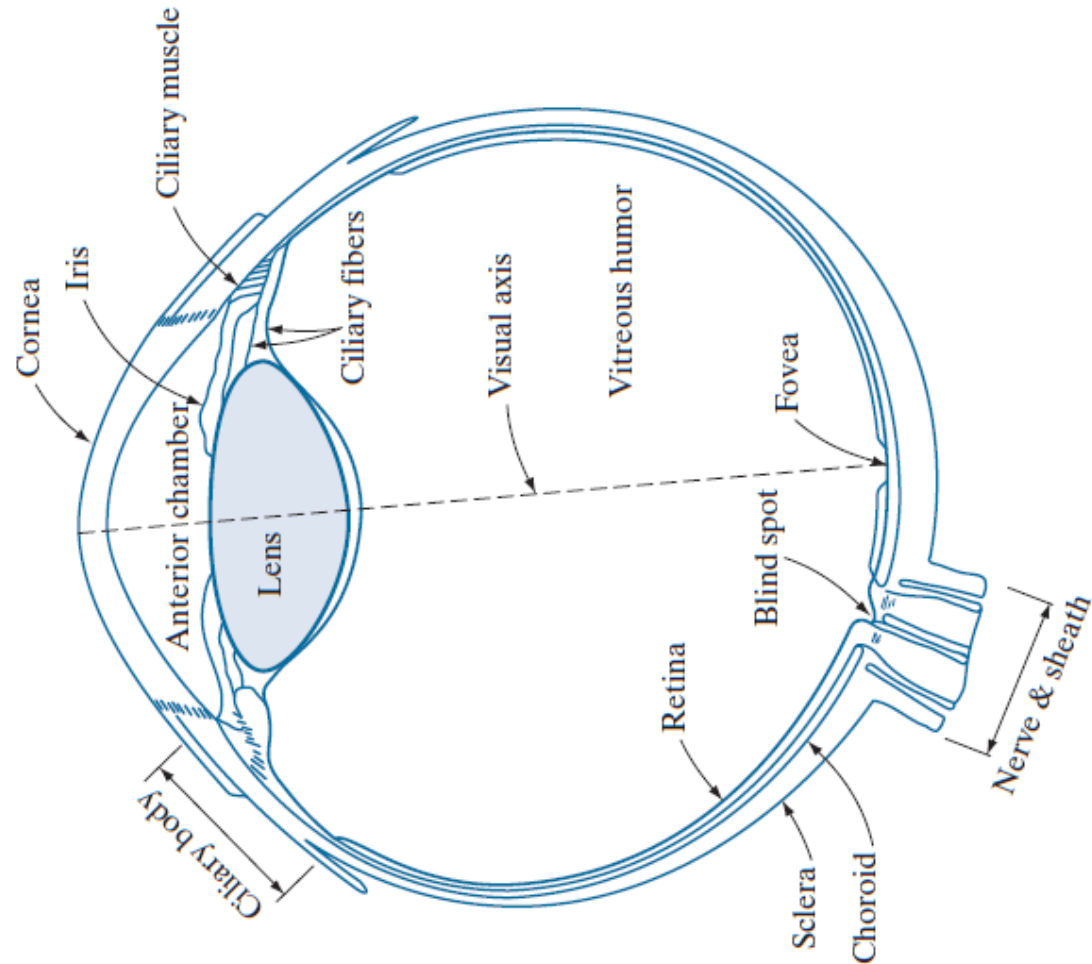


Pixel

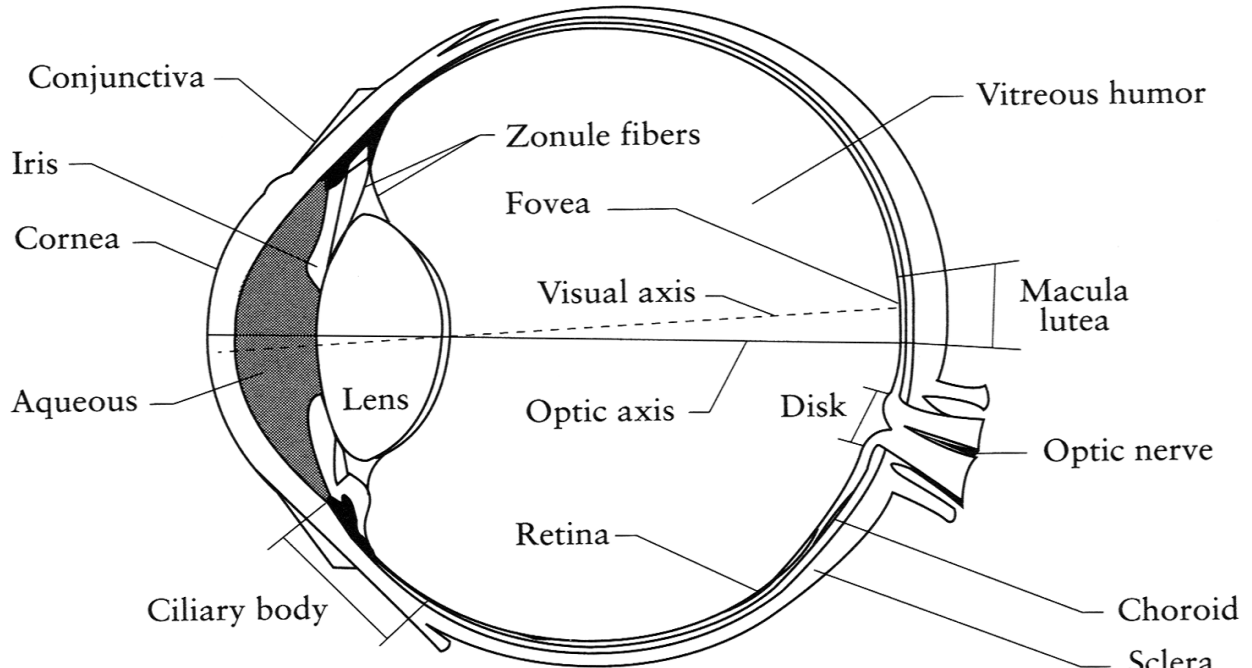
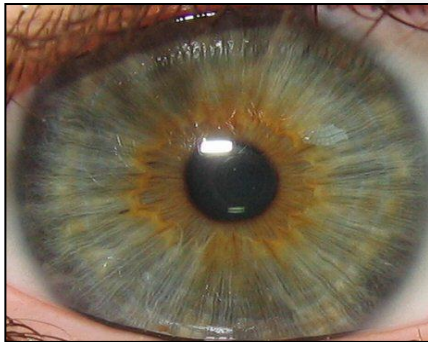
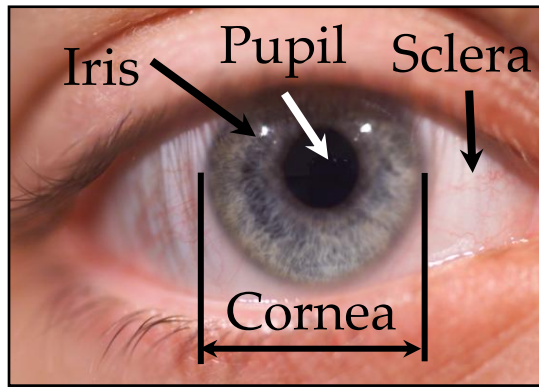
Pixel: A digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are referred to as *picture elements*, *image elements*, *pels*, and *pixels*.

Pixel is the term most widely used to denote the elements of a digital image

4. Visual Perception: Human Eye



Our Eyes



- Index of refraction: cornea 1.376, aqueous 1.336, lens 1.406-1.386
- Iris is the diaphragm that changes the aperture (pupil)
- Retina is the sensor where the fovea has the highest resolution

Contd...

Shape of Eye Nearly a sphere: Diameter of approximately 20mm

Three Membranes

- Cornea and Sclera outer cover
- Choroid
- Retina

Cornea: Tough, transparent, tissue that cover the anterior surface.

Sclera: Opaque membrane that encloses the remainder of the optic globe

Choroid: Lie directly below the sclera

- Contain a network of blood vessels
- Reduce the amount of extraneous light entering the eye
- Divided into the ciliary body and the iris diaphragm

Iris diaphragm: Contract or expand to control the amount of light (diameter 2-8 mm)

- The iris diaphragm controls amount of light that enters the eye.

Lens: Suspended by fibers attached to ciliary body

- Contains 60-70% water, 6% of fat.
- Colored by a slightly yellow pigmentation that increase with age

Visual Perception: Human Eye (cont.)

Pattern vision is afforded by the distribution of discrete light receptors over the surface of the retina.

Retina: The innermost membrane

Light: imaged on retina

There are two classes of receptors: **cones and rods**.

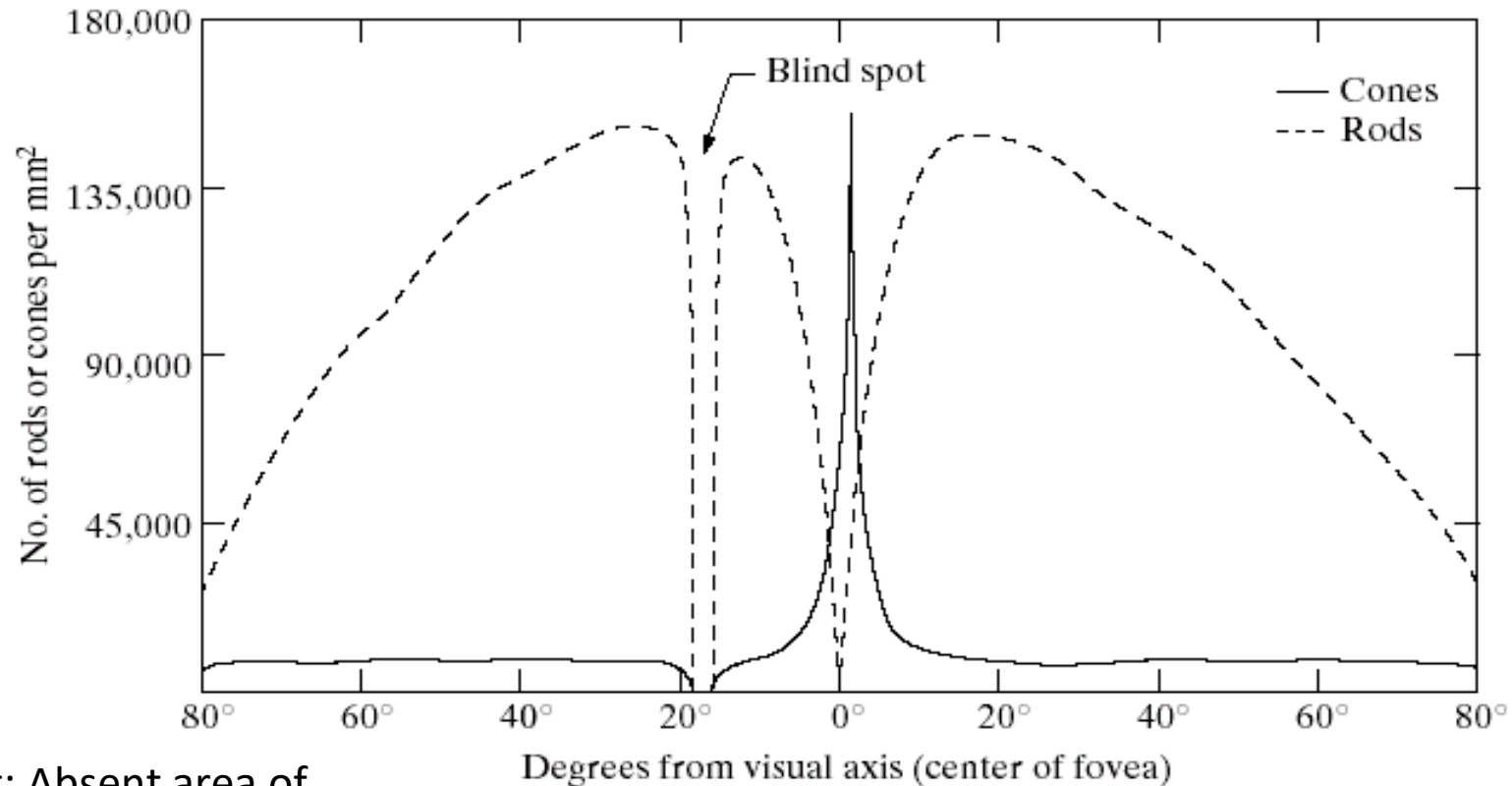
Cones (Photopic or bright-light vision)

- The number of cones in each eye: 6 to 7 millions
- The cones are concentrated in the central portion of the retina (fovea).
- Color receptors, high resolution in the fovea, less sensitive to light
- Related to chromic vision
- Resolve fine details

Rods : Scotopic (dim-light) vision

- The number of rods in each eye: 75 to 150 millions
- The rods are distributed over the retinal surface.
- Give a general, overall picture of the field of views color blind,
- Much more sensitive to light (night vision), lower resolution

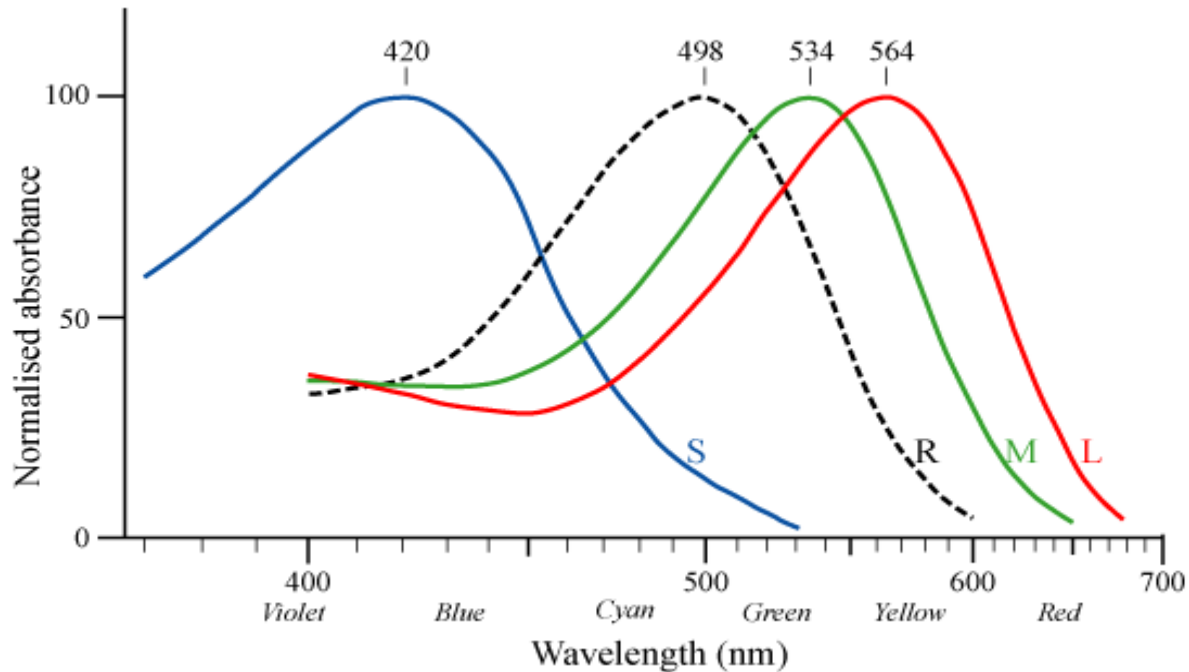
Distribution of Rods and Cones in the Retina



Blind Spot: Absent area of receptors

How do we sense color?

- Do we have infinite number of filters?



... rod

cones

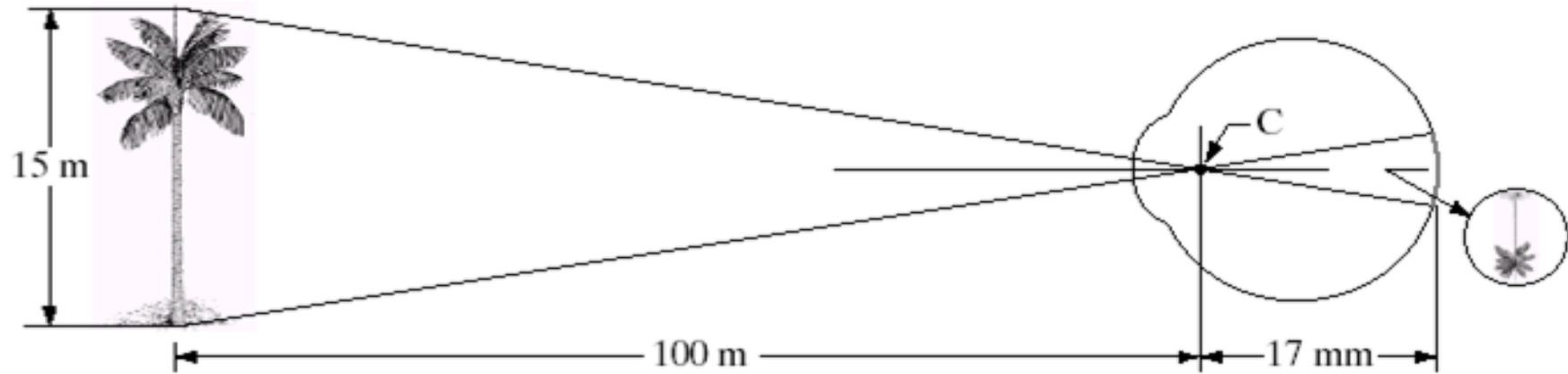
Three filters of different spectral responses

Blind Spot in Eye

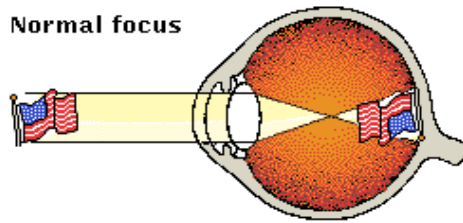


Close your right eye and look directly at the “+”

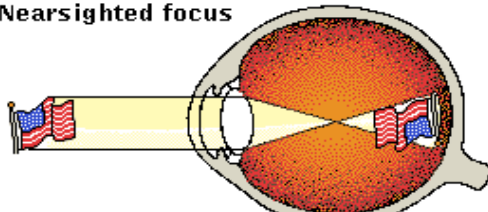
Image Formation in the Eye



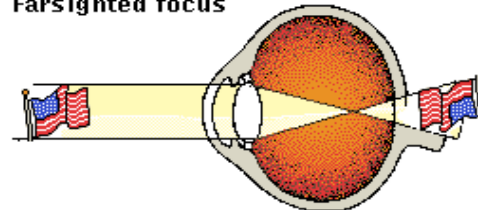
Normal focus



Nearsighted focus



Farsighted focus



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

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- Focal length of the eye: 17 to 14 mm
- Let h be the height in mm of that object in the retinal image, then the geometry of figure yields

$$15/100 = h / 17 , h = 2.55\text{mm}$$

- Retinal image is focused primarily on the region of the fovea
- Perception then takes place by the relative excitation of light receptors, which transforms radiant energy into electrical impulses that ultimately are decoded by the brain

Brightness adaption and Discrimination

- Digital images are displayed as a discrete set of intensities, the eyes ability to discriminate between different intensity levels is an important consideration in presenting image processing results.
- The range of brightness that the eye can adapt to is enormous, roughly on the order of 10^{10} - from the scotopic threshold to glare limit.
- Subjective Brightness (**intensity as perceived by the human visual system**) is logarithmic function of the light intensity incident on the eye. **on the order of 10^{10}**
- Photopic vision alone has a range of about 10^6 .
- Brightness adaptation: example “ B_a ”
- mL: millilambert

Brightness Adaption

- The essential point in interpreting the impressive dynamic range depicted in previous Figure is that the visual system cannot operate over such a range simultaneously.
- Rather, it accomplishes this large variation by changing its overall sensitivity, a phenomenon known as adaptive brightness.
- The total range of distinct intensity levels the eye can discriminate simultaneously is rather small when compared with the total adaption range.

- For any given set of conditions, the current sensitivity of the visual system is called brightness adaption level, which may correspond , for example, to brightness B_a in Figure.
- The short intersecting curve represents the range of subjective brightness that the eye can perceive when adapted to this level.
- The range is rather restricted, having a level B_b at an below which all stimuli are perceived as indistinguishable blacks.
- The upper portion of the curve is not actually restricted but, if extended too far, loses its meaning because much higher intensities would simple raise adaption level higher than B_a

Simultaneous Contrast

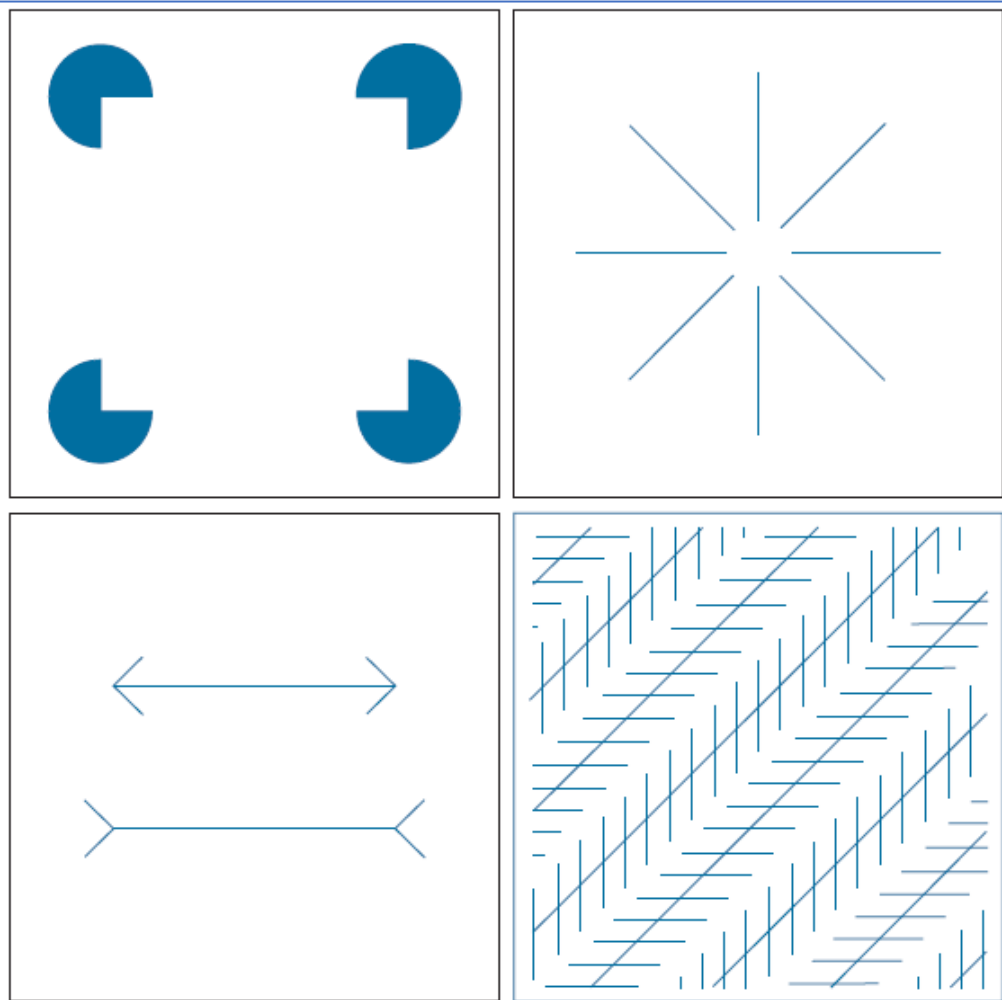
Simultaneous contrast: All small squares have exactly the same intensity but they appear progressively darker as background becomes lighter.



a b c

FIGURE Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

Examples for Human Perception Phenomena



a b
c d

FIGURE Some well-known optical illusions.

5. Light and the Electromagnetic Spectrum

The imaging machines can cover almost the entire EM spectrum, ranging from gamma to radio waves. These include

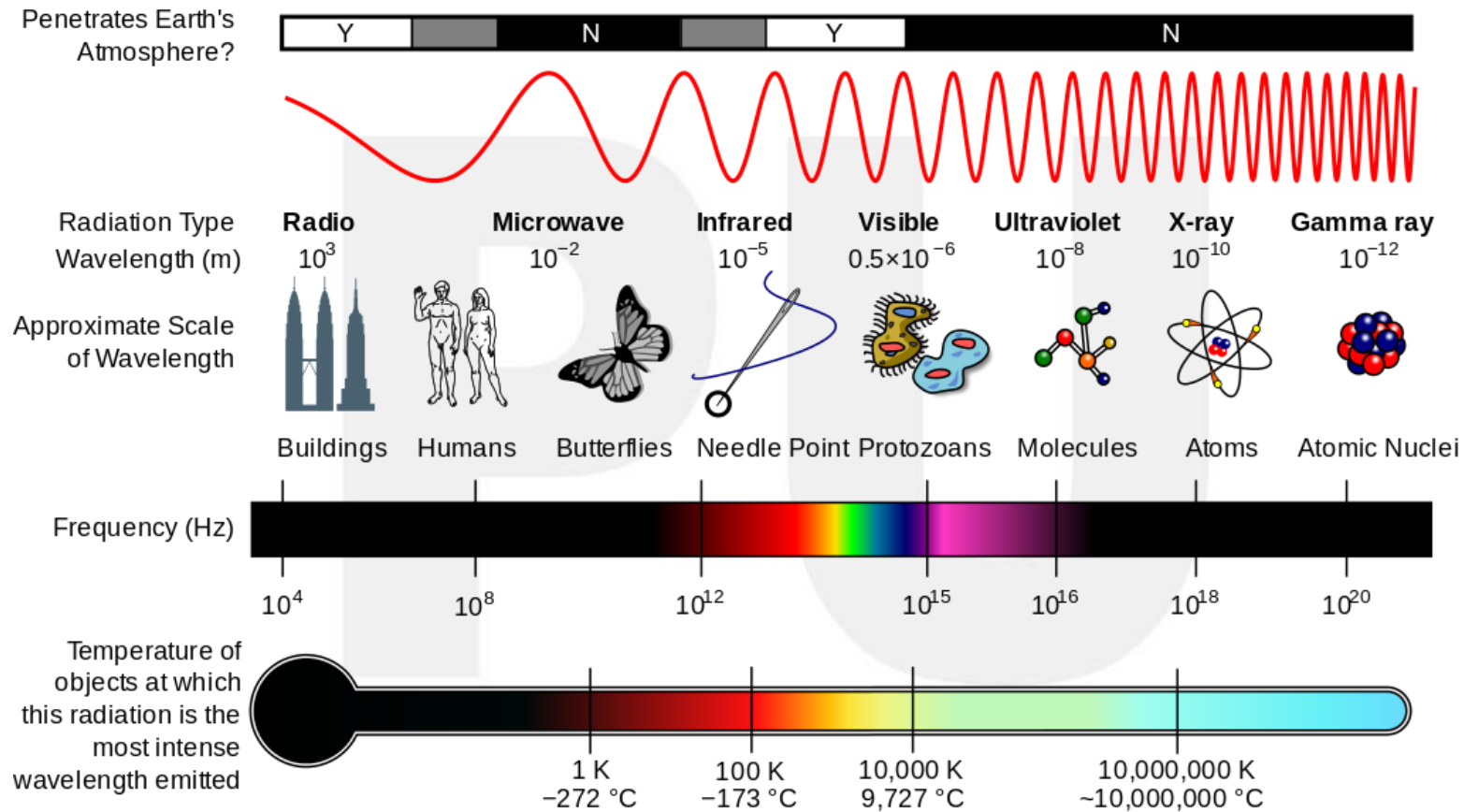
- Gamma ray images
- x-ray band images
- ultra-violet band images
- visual light and infra-red images
- Imaging based on micro-waves and radio waves

Non-EM band Imaging

- Acoustic and ultrasonic images (Geological application)
- Transmission Electron Microscopy
- Computer-generated synthetic images

Thus, digital image processing encompasses a wide and varied field of applications.

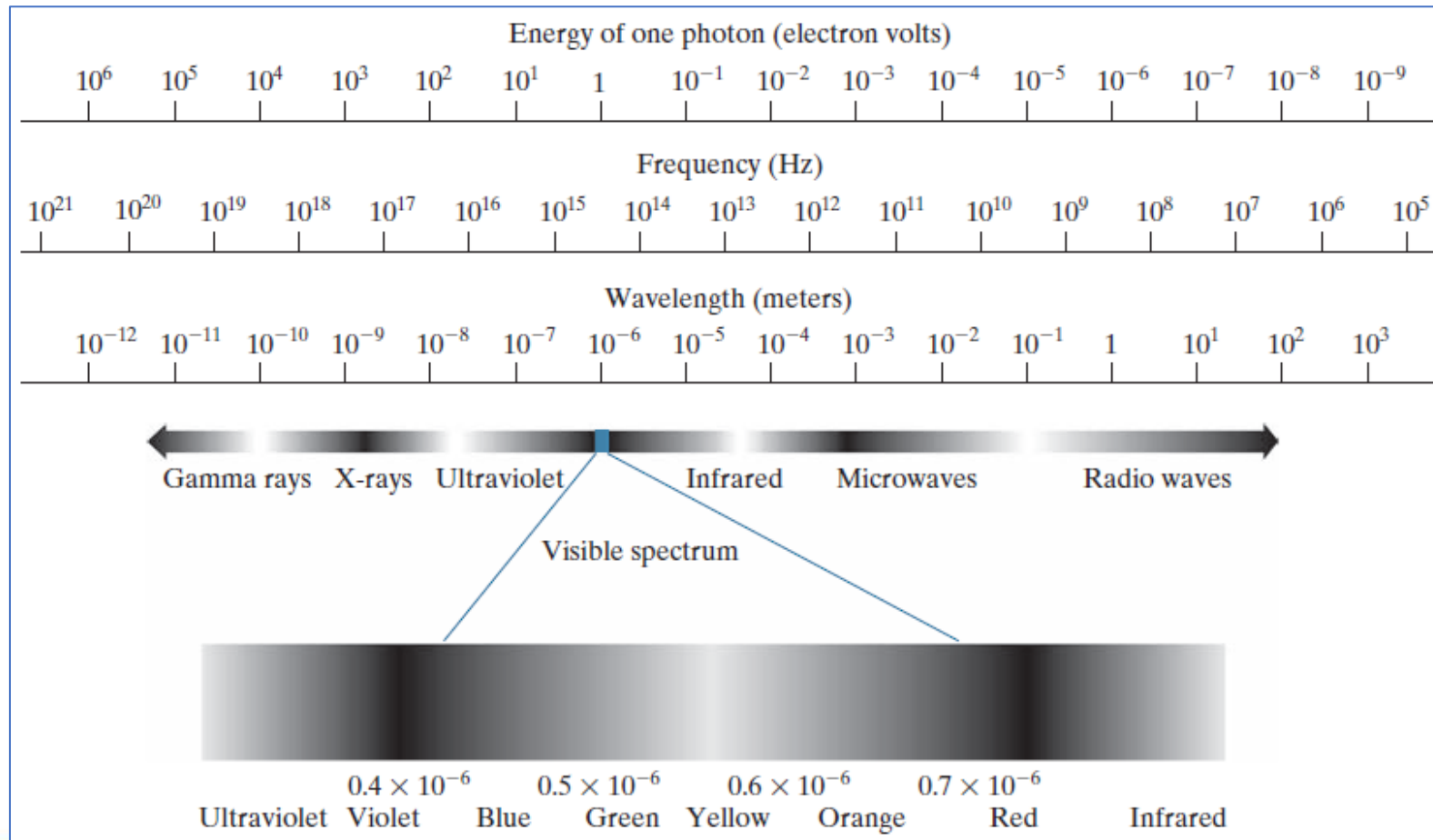
Light and the Electromagnetic Spectrum



Courtesy: https://en.wikipedia.org/wiki/Electromagnetic_spectrum#/media/File:EM_Spectrum_Properties_edit.svg

Contd....

FIGURE The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanations, but note that it encompasses a very narrow range of the total EM spectrum.



The colors that humans perceive in an object are determined by the nature of the light reflected from the object. i.e., green objects reflect light with wavelengths primarily in the 500 to 570 nm range while absorbing most of the energy at other wavelength

Monochromatic light: void of color

Intensity is the only attribute, from black to white

Monochromatic images are referred to as gray-scale images

Chromatic light bands: 0.43 to 0.79 μm

Three basic quantities described the quality of a chromatic light source:

Radiance: Total amount of energy

Luminance (lm): Amount of energy an observer perceives from a light source

Brightness: A subjective descriptor of light perception that is impossible to measure. It embodies the achromatic notion of intensity and one of the key factors in describing color sensation

- Relationship between frequency (ν) and wavelength (λ)

$$\lambda = \frac{c}{\nu}, \quad \text{where } c \text{ is the speed of light}$$

- Energy of a photon

$$E = h\nu, \quad \text{where } h \text{ is Planck's constant}$$

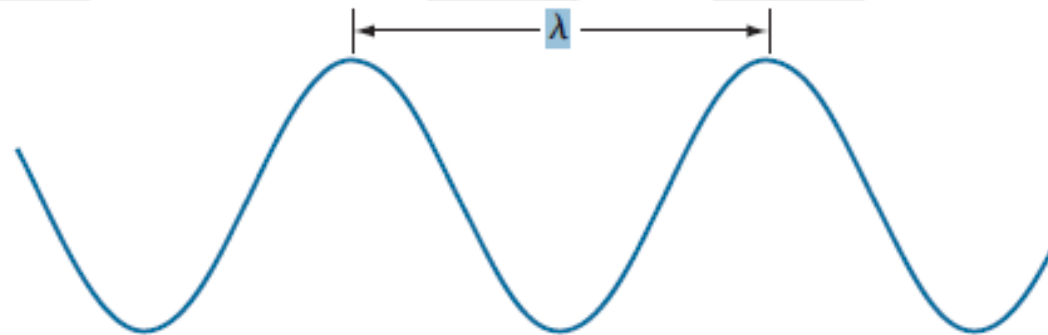
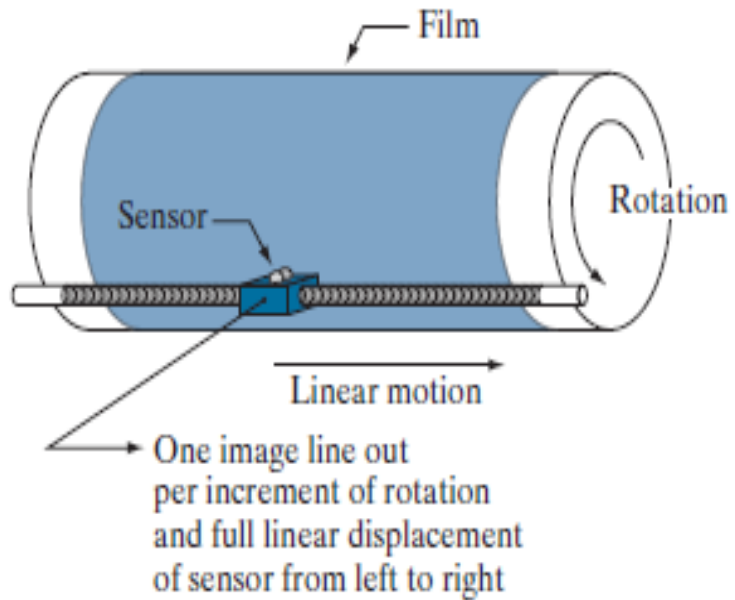


FIGURE Graphical representation of one wavelength.

6. Image Acquisition Using a Single Sensor



FIGURE

Combining a single sensing element with mechanical motion to generate a 2-D image.

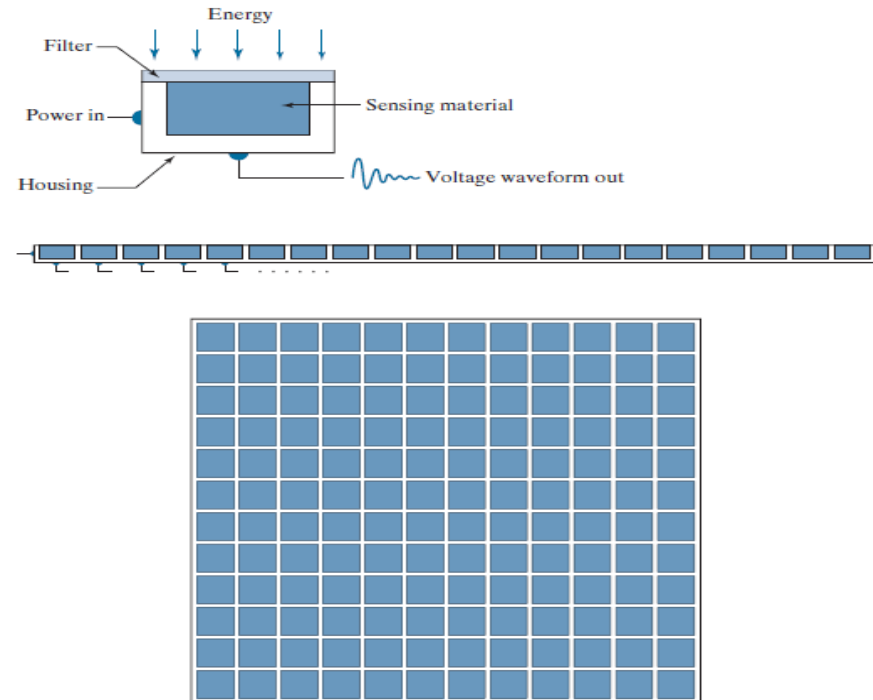
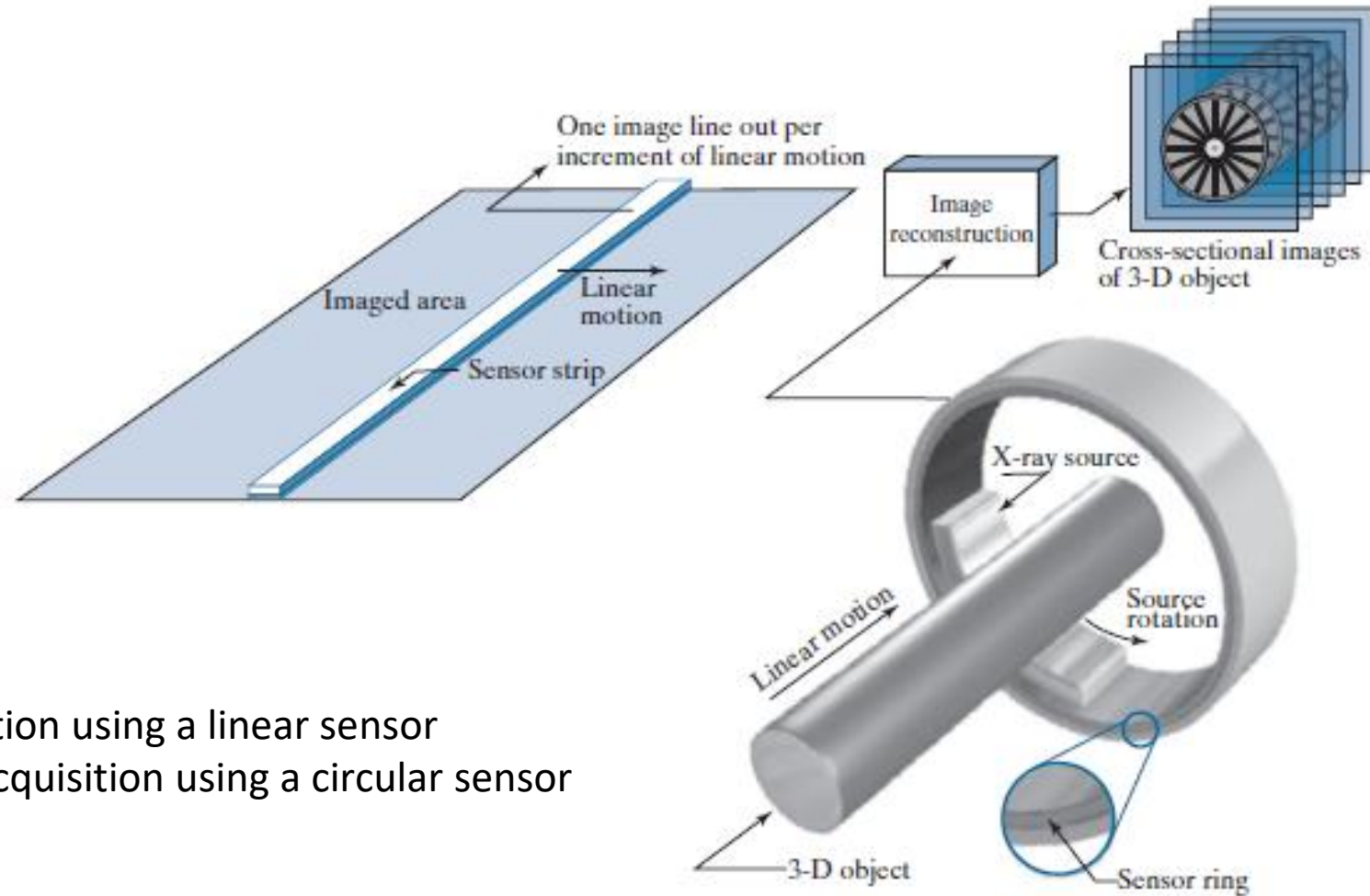


FIGURE (a) Single sensing element. (b) Line sensor. (c) Array sensor.

Image Acquisition Using Sensor Strips



Fingerprint sweep sensor



FIGURE

(a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

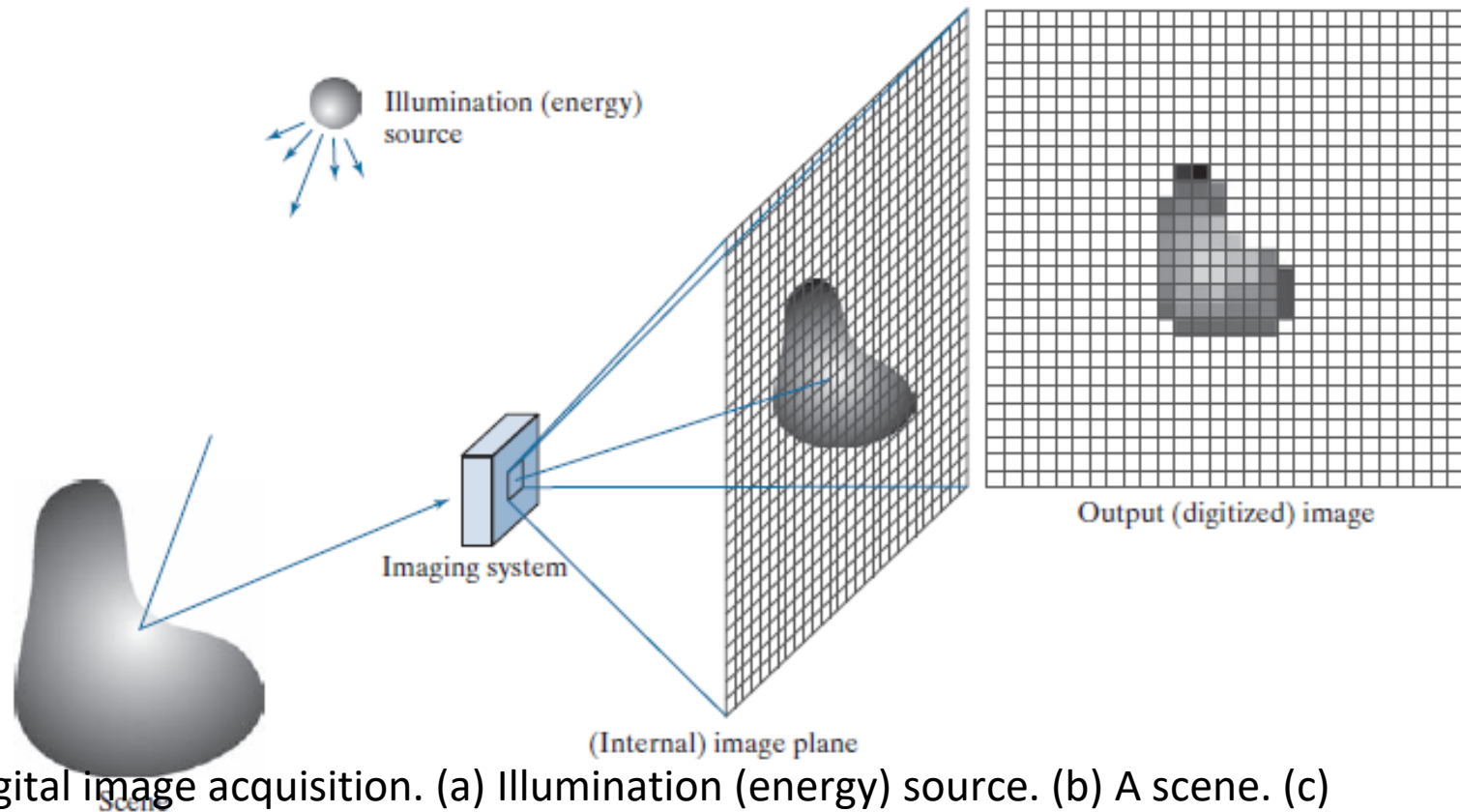


FIGURE An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

A Simple Image Formation Model

Let $f(x,y)$ be an image function, then $f(x,y) = i(x,y) r(x,y)$,

where $i(x,y)$: the illumination function

$r(x,y)$: the reflection function

Note: $0 < i(x,y) < \infty$ and $0 < r(x,y) < 1$.

For digital images the minimum gray level is usually 0, but the maximum depends on number of quantization levels used to digitize an image. The most common is 256 levels, so that the maximum level is 255.

Binary images: images having only two possible brightness levels (black and white)

Gray scale images: “black and white” images

Color images: can be described mathematically as three gray scale images

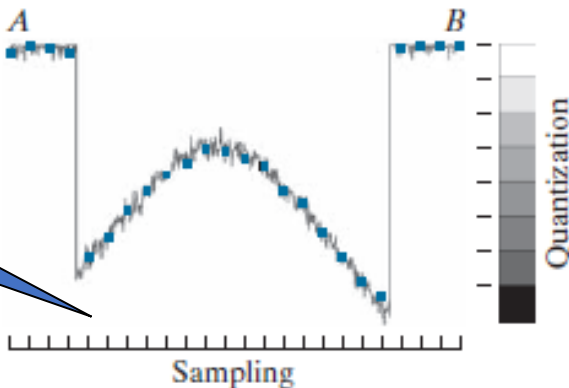
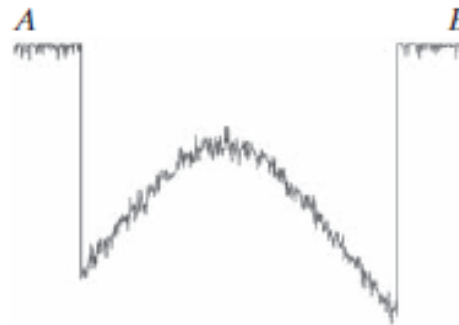
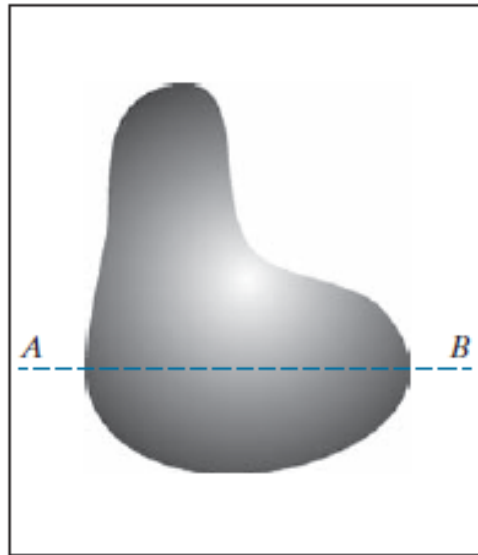
A Digital Image



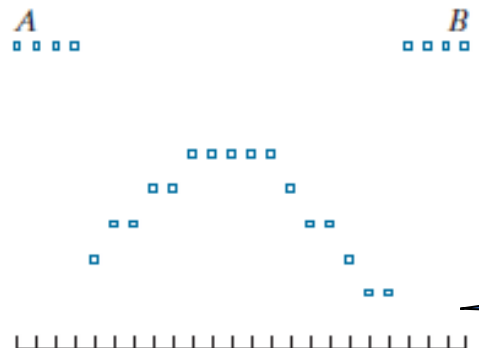
Image “After snow storm”

- **An image: a multidimensional function of spatial coordinates.**
- **Spatial coordinate:** (x,y) for 2D case such as photograph,
 (x,y,z) for 3D case such as CT scan images
 (x,y,t) for movies
- The function f may represent intensity (for monochrome images or color (for color images) or other associated values.

7. Image Sampling and Quantization



Digitizing the
coordinate
values



Digitizing the
amplitude values

a b

c d

FIGURE

(a) Continuous image. (b) A scan line showing intensity variations along line *AB* in the continuous image.

(c) Sampling and quantization.

(d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

Sampling: Digitization of the spatial coordinates (x,y)

Quantization: Digitization in amplitude (also called *gray-level quantization*)

8 bit quantization: $2^8 = 256$ gray levels (0: black, 255: white)

Binary (1 bit quantization): 2 gray levels (0: black, 1: white)

Commonly used number of samples (resolution)

Digital still cameras: 640x480, 1024x1024, up to 4064 x 2704

Digital video cameras: 640x480 at 30 frames/second
1920x1080 at 60 f/s (HDTV)

Image Sampling and Quantization

Sampling: digitizing the 2-dimensional spatial coordinate values

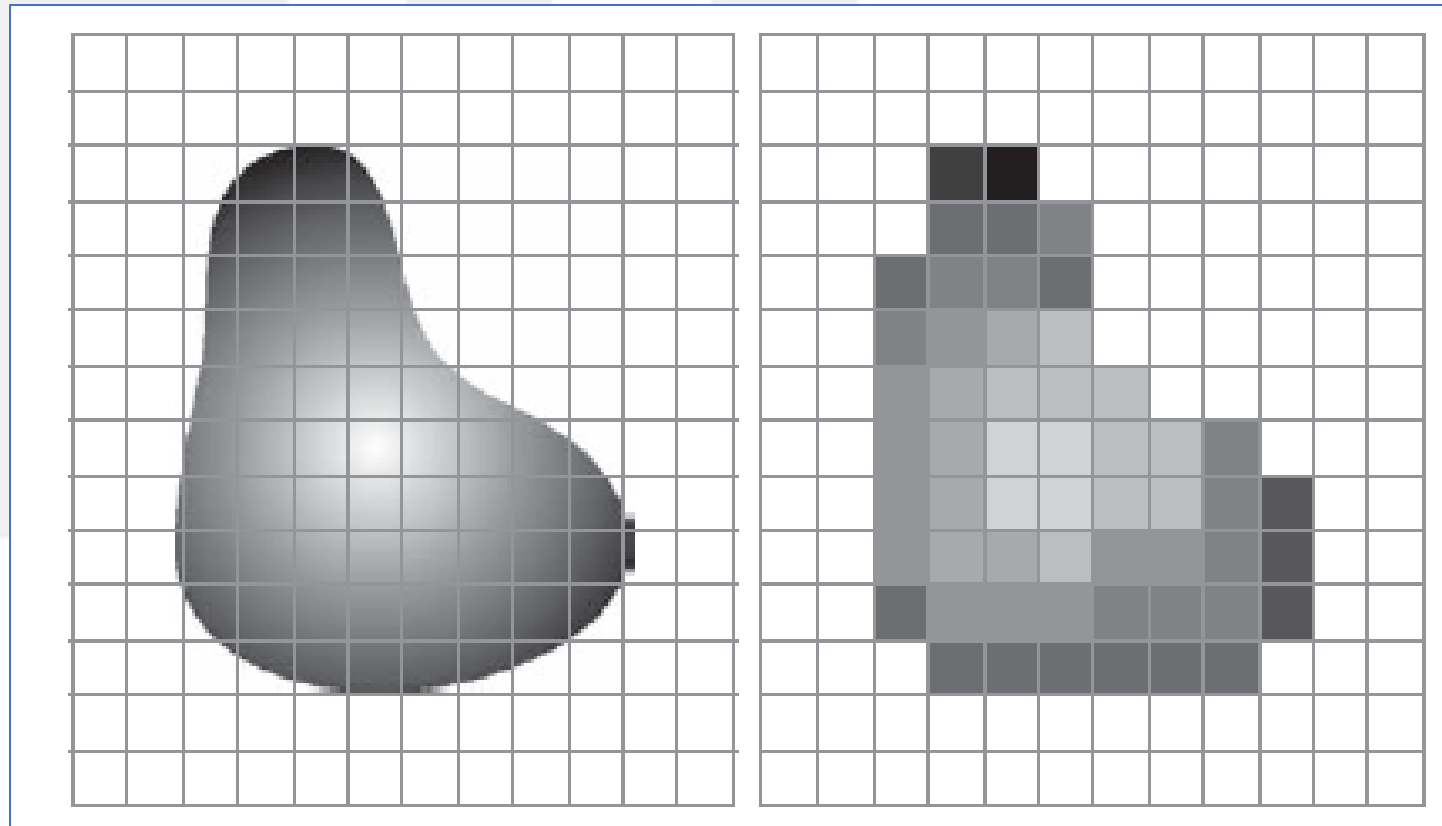
Quantization: digitizing the amplitude values (brightness level)

a b

FIGURE

(a) Continuous image projected onto a sensor array.

(b) Result of image sampling and quantization.



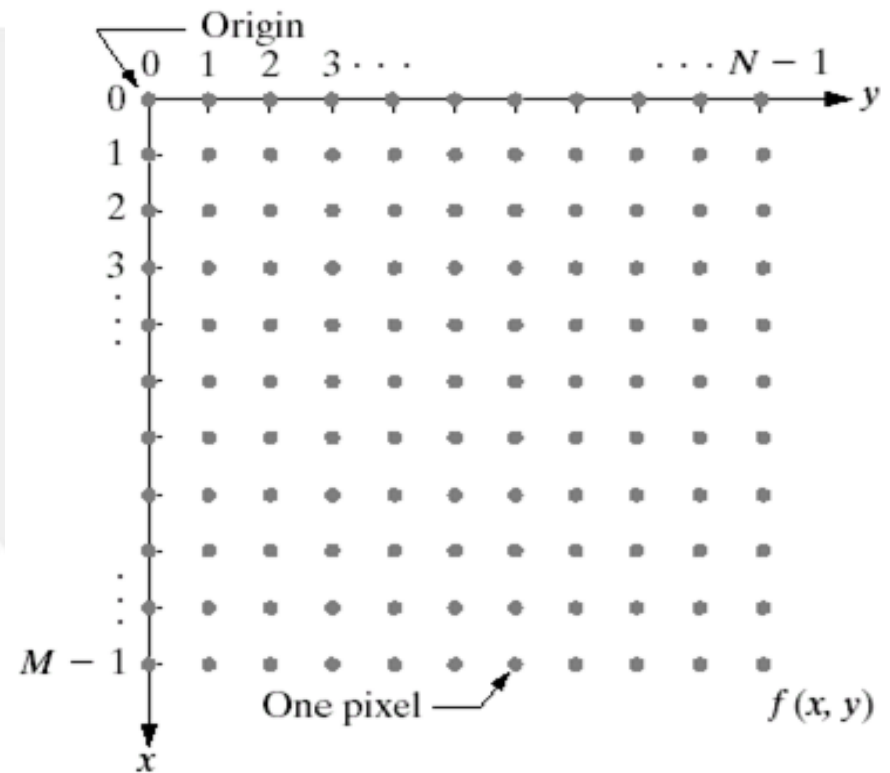
Sampling and Quantization

An $M \times N$ digital image is expressed as

$$\begin{array}{c} \text{Rows} \downarrow \end{array} \begin{array}{c} \xrightarrow{\text{Columns}} \\ \left[\begin{array}{cccc} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{array} \right] \end{array}$$

N : No of Columns

M : No of Rows



Representing Digital Images

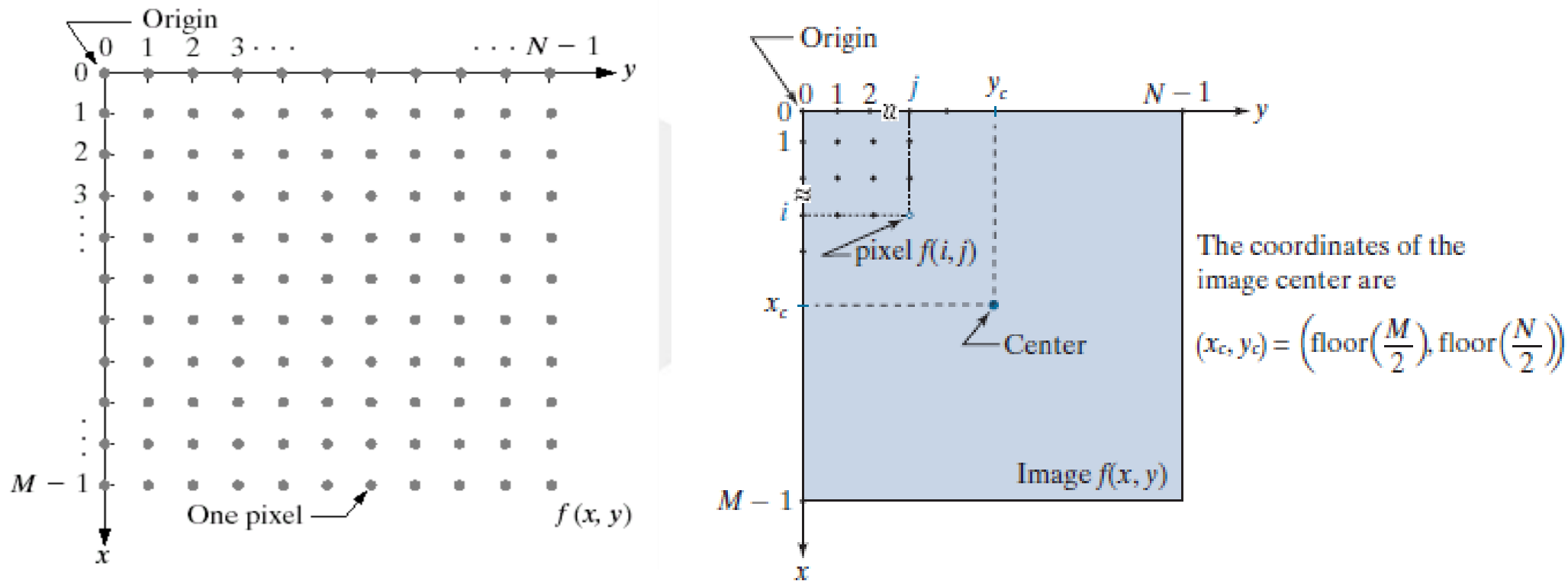


FIGURE Coordinate convention used to represent digital images. Because coordinate values are integers, there is a one-to-one correspondence between x and y and the rows (r) and columns (c) of a matrix.

Representing Digital Images

The representation of an $M \times N$ numerical array as

PU

Representing Digital Images

The representation of an $M \times N$ numerical array as

PU

Representing Digital Images

The representation of an $M \times N$ numerical array in MATLAB

PU

Representing Digital Images

Discrete intensity interval $[0, L-1]$, $L=2^k$

The number b of bits required to store
a $M \times N$ digitized image

$$b = M \times N \times m$$

For $N=M= 128$, $G=64$ ($m=6$)

$b= 98,304$

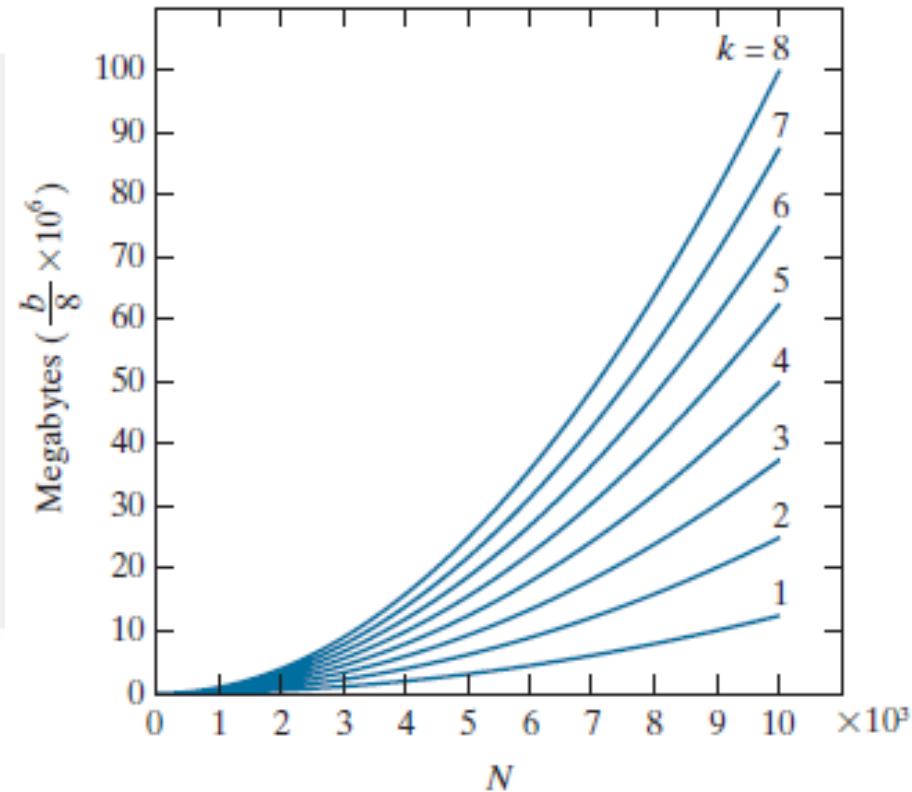


FIGURE Number of megabytes required to store images for various values of N and k .

Representing Digital Images

Table 1. Number of Storage bits for various values of N and K

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

- **Dynamic range:** The range of values spanned by the gray scale is referred to as the *dynamic range*, a term used in different ways in different fields. **The dynamic range of an imaging system to be the ratio of the maximum measurable intensity to the minimum detectable intensity level in the system.**
- The upper limit is determined by *saturation* and the lower limit by *noise*, although noise can be present also in lighter intensities. **The dynamic range establishes the lowest and highest intensity levels that a system can represent and, consequently, that an image can have.**
- **Image contrast:** The difference in intensity between the highest and lowest intensity levels in an image.
- The **contrast ratio** is the ratio of these two quantities.
- When an appreciable number of pixels in an image have a **high dynamic range**, we can expect the image to **have high contrast**. Conversely, an image with **low dynamic range** typically has a **dull, washed-out gray look**.

Sampling and Spatial Resolution

- **Spatial Aliasing Problem:** Spatial aliasing is insufficient sampling of data along the space axis, which occurs because of the insufficient spatial resolution of the acquired image.
- It is recommended to sample the image at a rate close to the ideal sampling rate (Nyquist Rate). Images obtained with sampling distances larger than those established by this rate suffer from undersampling.
- The Nyquist Rate is defined as twice the bandwidth of the continuous-time signal. It should be noted that the sampling frequency must be strictly greater than the Nyquist Rate of the signal to achieve unambiguous representation of the signal.
- The Critical Sampling Distance is the Sampling Distance corresponding to the Nyquist Rate.

Intensity resolution and Image Quantization

Intensity resolution : The smallest discernible change in intensity level stated with 8 bits, 10 bits, 12 bits, 16 bits, etc. Intensity quantization using 32 bits is rare.

For example it is common to say that an image whose intensity is quantized into 256 levels has 8 bits of intensity resolution.

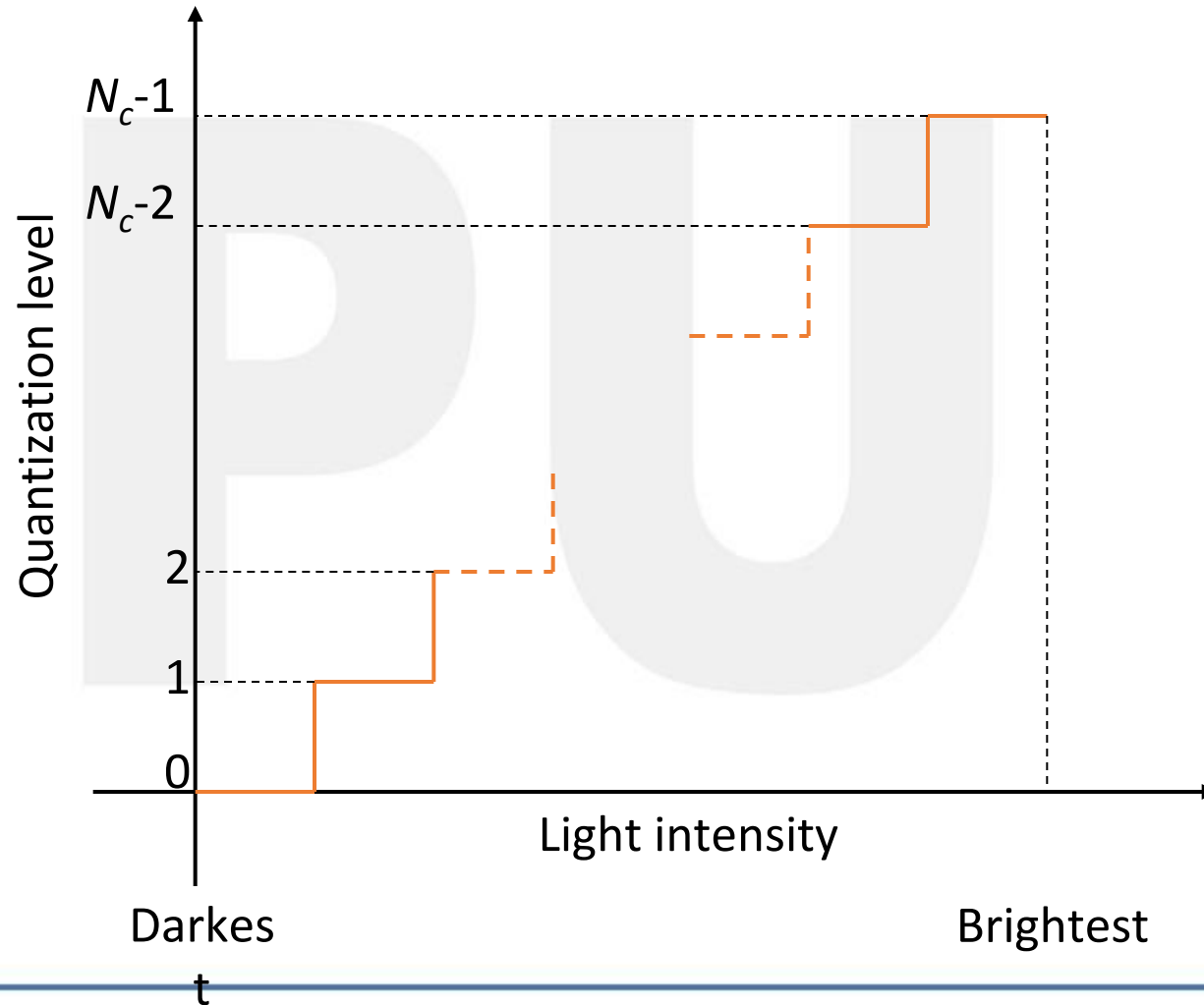
Image quantization: discretize continuous pixel values into discrete numbers

Color resolution/ color depth/ levels:

- No. of colors or gray levels or
- No. of bits representing each pixel value
- No. of colors or gray levels N_c is given by
- where b = no. of bits

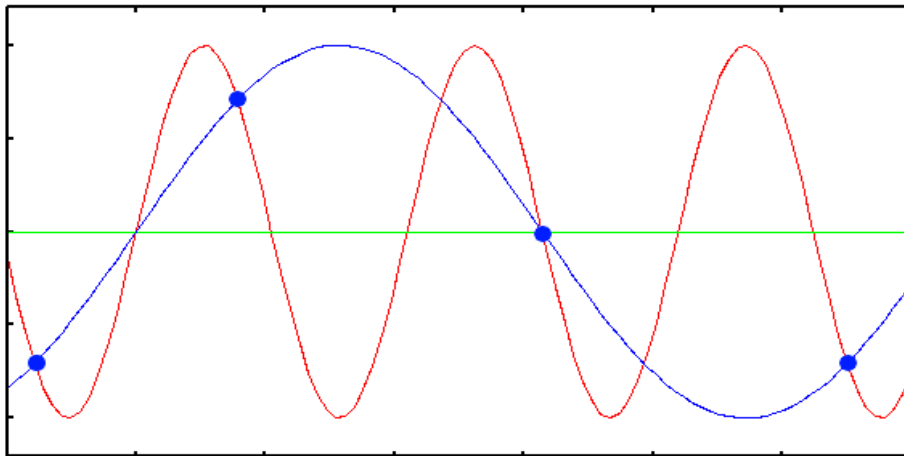
$$N_c = 2^b$$

Quantization function



Aliasing

Sampling and Spatial Resolution Spatial Aliasing Problem: In statistics, signal processing, and related disciplines, aliasing is an effect that causes different continuous signals to become indistinguishable (or aliases of one another) when sampled. When this happens, the original signal cannot be uniquely reconstructed from the sampled signal.

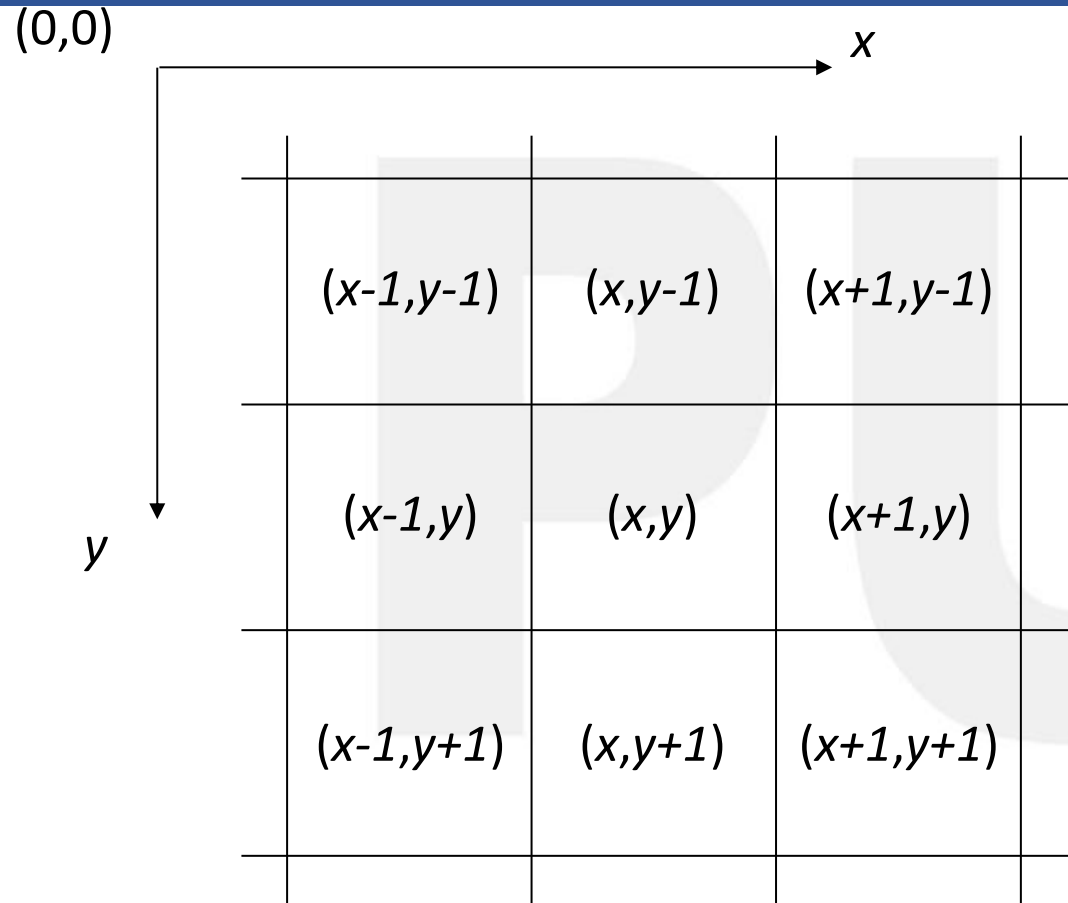


Aliasing of a *undersampled 1D sinusoidal signal*



Aliasing of a *undersampled 2D image*

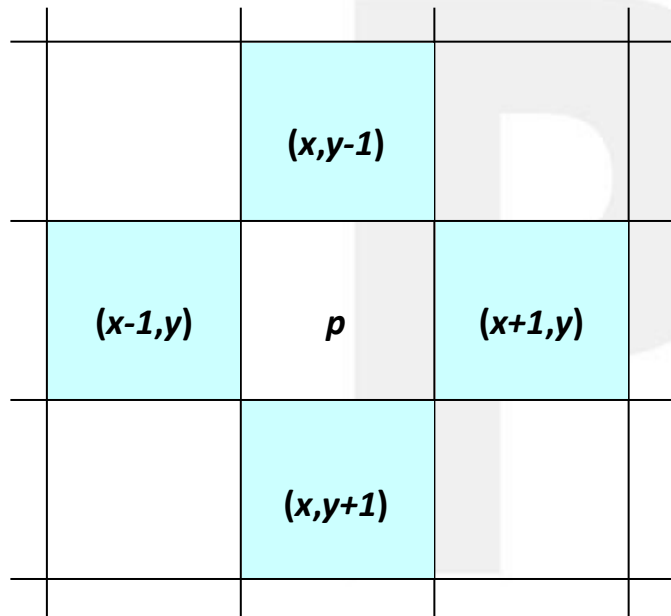
8. Some Basic Relationship of Pixels



Conventional indexing method

Neighbors of a Pixel

Neighborhood relation is used to tell adjacent pixels. It is useful for analyzing regions.



4-neighbors of p :

$$N_4(p) = \left\{ \begin{array}{l} (x-1,y) \\ (x+1,y) \\ (x,y-1) \\ (x,y+1) \end{array} \right\}$$

4-neighborhood relation considers only vertical and horizontal neighbors.

Note: $q \in N_4(p)$ implies $p \in N_4(q)$

Contd...

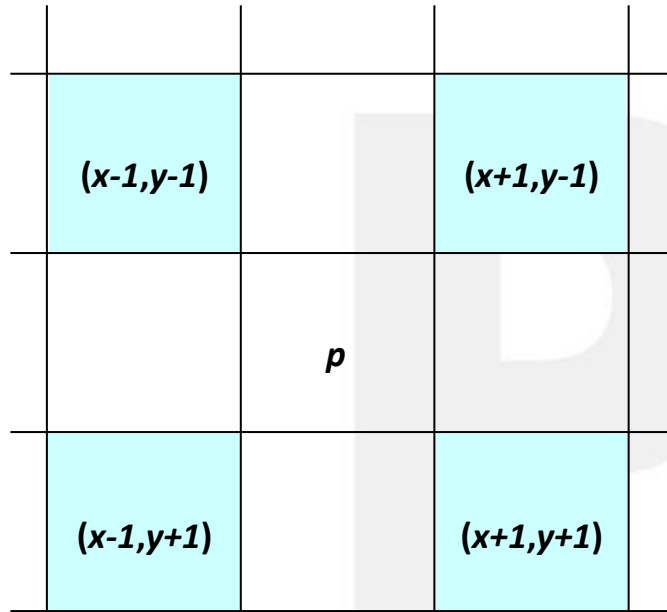
	$(x-1, y-1)$	$(x, y-1)$	$(x+1, y-1)$
	$(x-1, y)$	p	$(x+1, y)$
	$(x-1, y+1)$	$(x, y+1)$	$(x+1, y+1)$

8-neighbors of p :

$$N_8(p) = \left\{ \begin{array}{l} (x-1, y-1) \\ (x, y-1) \\ (x+1, y-1) \\ (x-1, y) \\ (x+1, y) \\ (x-1, y+1) \\ (x, y+1) \\ (x+1, y+1) \end{array} \right\}$$

8-neighborhood relation considers all neighbor pixels.

Contd...



Diagonal neighbors of p :

$$N_D(p) = \left\{ \begin{array}{l} (x-1, y-1) \\ (x+1, y-1) \\ (x-1, y+1) \\ (x+1, y+1) \end{array} \right\}$$

Diagonal -neighborhood relation considers only diagonal neighbor pixels.

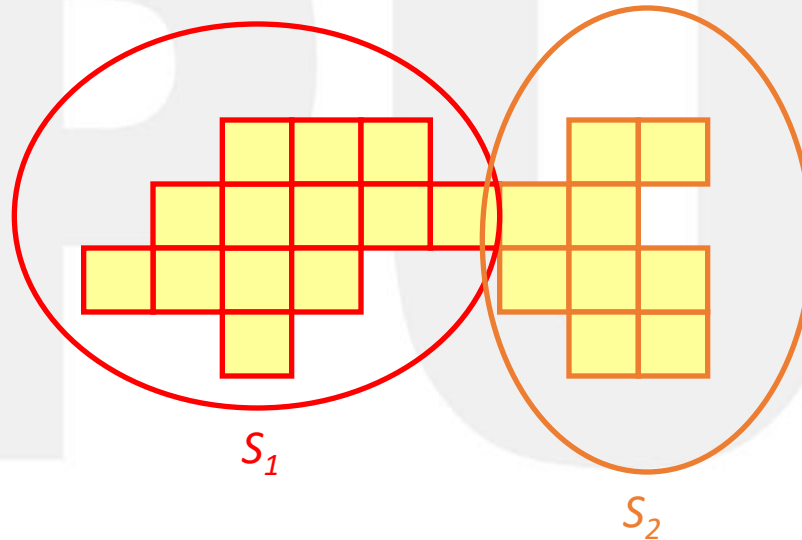
Adjacency and Connectivity

- Let V : a set of intensity values used to define adjacency and connectivity.
- In a binary image, $V = \{1\}$, if we are referring to adjacency of pixels with value 1.
- In a gray-scale image, the idea is the same, but V typically contains more elements, for example, $V = \{180, 181, 182, \dots, 200\}$
- If the possible intensity values 0 – 255, V set can be any subset of these 256 values.

Adjacency

A pixel p is *adjacent* to pixel q if they are connected.

Two image subsets S_1 and S_2 are adjacent if some pixel in S_1 is adjacent to some pixel in S_2 .



We can define type of adjacency: **4-adjacency**, **8-adjacency** or **m-adjacency** depending on type of connectivity.

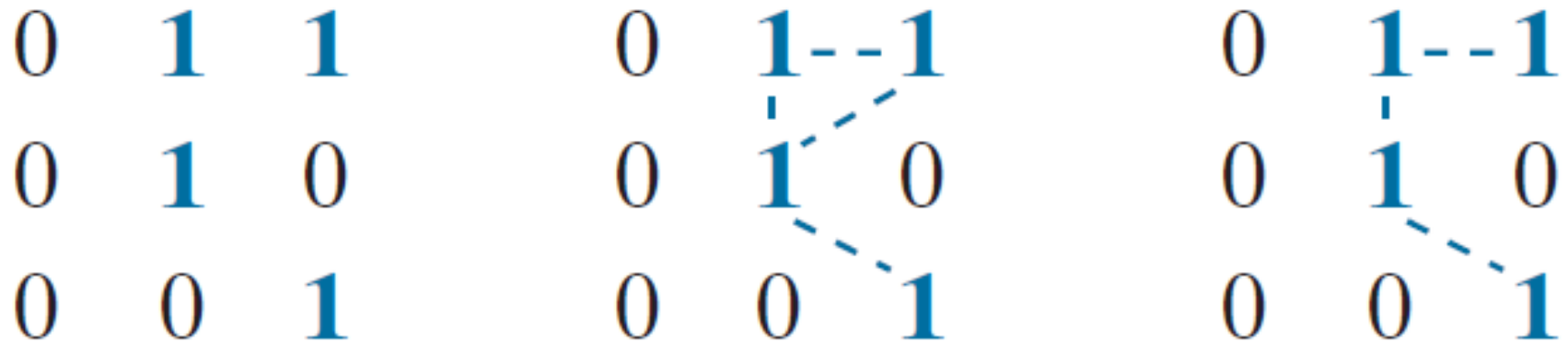
Types of Adjacency

1. **4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.
2. **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.
3. **m-adjacency (mixed):** Two pixels p and q with values from V are m-adjacent if :
 - q is in $N_4(p)$ **or**
 - q is in $N_D(p)$ **and** the set $N_4(p) \cap N_4(q)$ has no pixel whose values are from V (no intersection)
 - i.e., $N_4(p) \cap N_4(q) = \emptyset$

Important Note: the type of adjacency used must be specified

Types of Adjacency

- Mixed adjacency is a modification of 8-adjacency. It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.
- For example:



a b c

FIGURE (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) *m*-adjacency.

Types of Adjacency

- In this example, we can note that to connect between two pixels (finding a path between two pixels):
 - In 8-adjacency way, you can find multiple paths between two pixels
 - While, in m-adjacency, you can find only one path between two pixels
- So, m-adjacency has eliminated the multiple path connection that has been generated by the 8-adjacency.
- Two subsets $S1$ and $S2$ are adjacent, if some pixel in $S1$ is adjacent to some pixel in $S2$. Adjacent means, either 4-, 8- or m-adjacency.

A Digital Path

A digital path (or curve) from pixel p with coordinate (x,y) to pixel q with coordinate (s,t) is a sequence of distinct pixels with coordinates $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ where $(x_0, y_0) = (x, y)$ and $(x_n, y_n) = (s, t)$ and pixels (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \leq i \leq n$

- n is the length of the path
- If $(x_0, y_0) = (x_n, y_n)$, the path is closed.
- We can specify 4-, 8- or m -paths depending on the type of adjacency specified.

Path

A **path** from pixel p at (x,y) to pixel q at (s,t) is a sequence of distinct pixels:

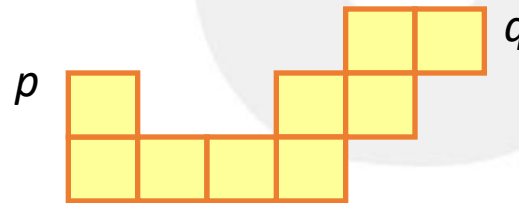
$$(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

such that

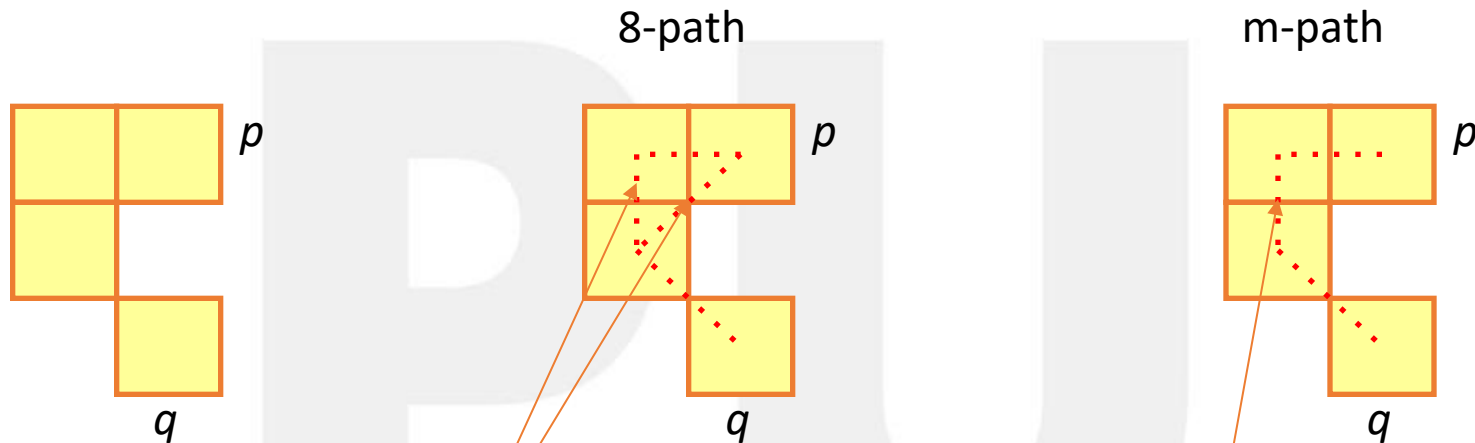
$$(x_0, y_0) = (x, y) \text{ and } (x_n, y_n) = (s, t)$$

and

$$(x_i, y_i) \text{ is adjacent to } (x_{i-1}, y_{i-1}), \quad i = 1, \dots, n$$



We can define type of path: 4-path, 8-path or m-path depending on type of adjacency.



8-path from p to q
results in some ambiguity

m-path from p to q
solves this ambiguity

Connectivity

- Let S represent a subset of pixels in an image, two pixels p and q are said to be connected in S if there exists a path between them consisting entirely of pixels in S .
- For any pixel p in S , the set of pixels that are connected to it in S is called a *connected component* of S . If it only has one connected component, then set S is called a *connected set*.

Region and Boundary

- **Region**

Let R be a subset of pixels in an image, we call R a region of the image if R is a connected set.

- **Boundary**

The *boundary* (also called *border* or *contour*) of a region R is the set of pixels in the region that have one or more neighbors that are not in R .

Region and Boundary

- If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns in the image.
- This extra definition is required because an image has no neighbors beyond its borders
- Normally, when we refer to a region, we are referring to subset of an image, and any pixels in the boundary of the region that happen to coincide with the border of the image are included implicitly as part of the region boundary.

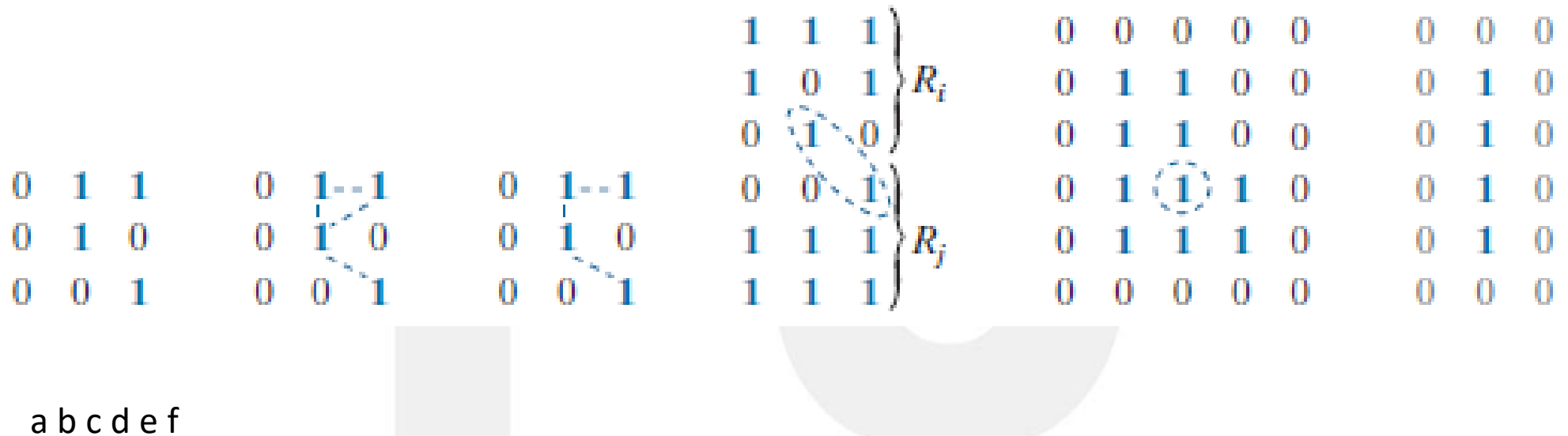


FIGURE (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines). (c) m -adjacency. (d) Two regions (of 1's) that are 8-adjacent. (e) The circled point is on the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

Distance Measures

- For pixels p , q and z , with coordinates (x,y) , (s,t) and (v,w) , respectively, D is a distance function if:
 - (a) $D(p,q) \geq 0$ ($D(p,q) = 0$ iff $p = q$),
 - (b) $D(p,q) = D(q,p)$, and
 - (c) $D(p,z) \leq D(p,q) + D(q,z)$.

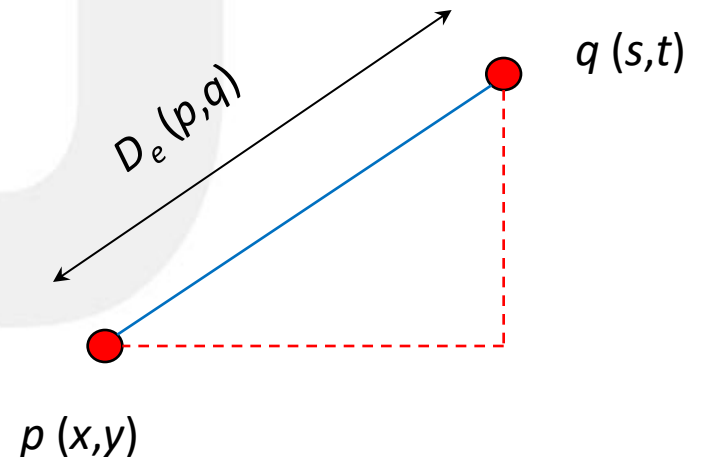
Distance Measures

- The *Euclidean Distance* between p and q is defined as:

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$$

Pixels having a distance less than or equal to some value r from (x, y) are the points contained in a disk of radius r centered at (x, y)

$$D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$

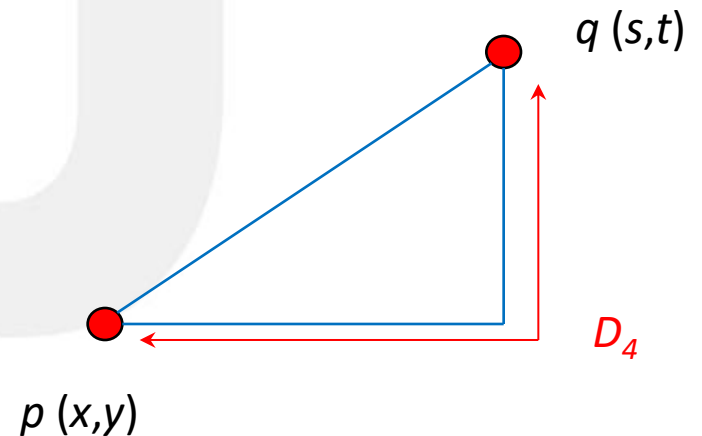


Distance Measures

The D_4 distance (also called *city-block distance*) between p and q is defined as:

$$D_4(p, q) = |x - s| + |y - t|$$

Pixels having a D_4 distance from (x, y) , less than or equal to some value r form a Diamond centered at (x, y)

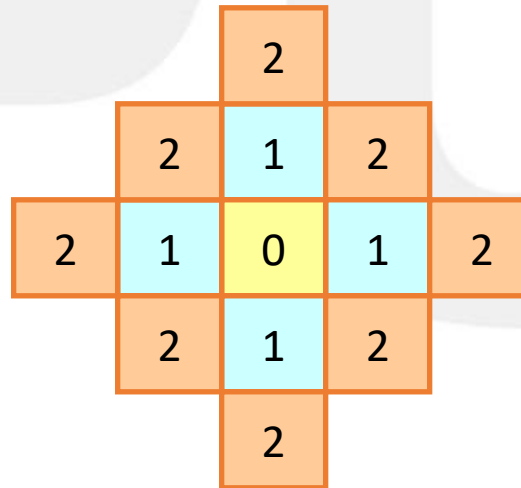


$$D_4(p, q) = |x - s| + |y - t|$$

Distance Measures

Example: The pixels with distance $D_4 \leq 2$ from (x,y) form the following contours of constant distance.

The pixels with $D_4 = 1$ are the 4-neighbors of (x,y)



Distance Measures

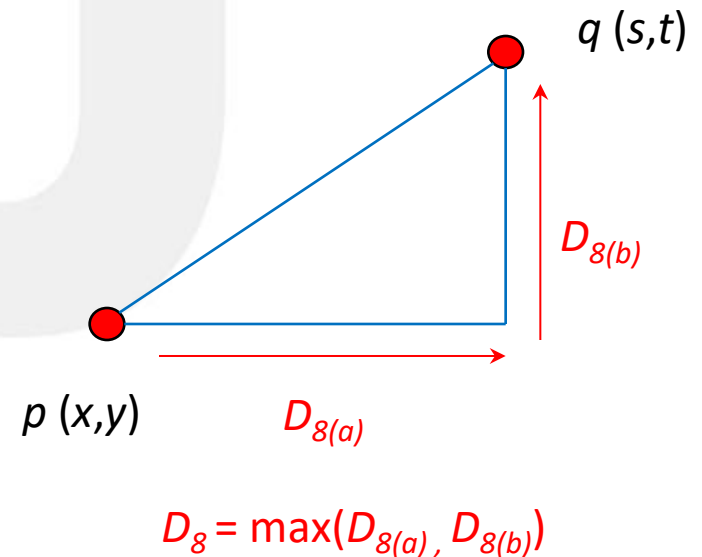
The D_8 distance (also called *chessboard distance*) between p and q is defined as:

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

Pixels having a D_8 distance from (x, y) , less than or equal to some value r form a square

Centered at (x, y)

$$D_8(p, q) = \max(|x - s|, |y - t|)$$



Distance Measures

Example: D_g distance ≤ 2 from (x,y) form the following contours of constant distance.

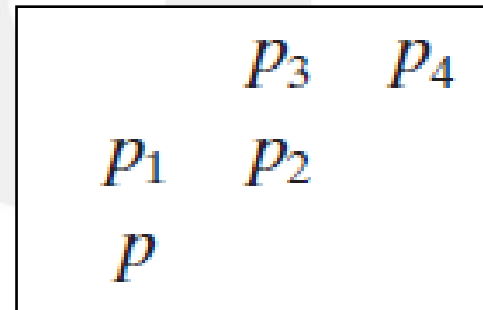
2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Distance Measures

- **Dm distance:** is defined as the shortest m-path between the points. In this case, the distance between two pixels will depend on the values of the pixels along the path, as well as the values of their neighbors.

Example: Consider the following arrangement of pixels and assume that p , p_2 , and p_4 have value 1 and that p_1 and p_3 can have a value of 0 or 1.

Suppose that we consider the adjacency of pixels values 1 (i.e. $V = \{1\}$)



Distance Measures

Cont. Example:

Now, to compute the D_m between points p and p_4

Here we have 4 cases:

Case1: If $p_1 = 0$ and $p_3 = 0$

The length of the shortest m-path
(the D_m distance) is 2 (p, p_2, p_4)

	0	1
0	1	
1		

Distance Measures

Cont. Example:

Case2: If $p_1 = 1$ and $p_3 = 0$

now, p_1 and p will no longer be adjacent (see m -adjacency definition)

then, the length of the shortest path will be 3 (p, p_1, p_2, p_4)

	0	1
1	1	
1		

Distance Measures

Cont. Example:

Case3: If $p_1 = 0$ and $p_3 = 1$

The same applies here, and the shortest –m-path will be 3 (p, p_2, p_3, p_4)

	1	1
0	1	
1		

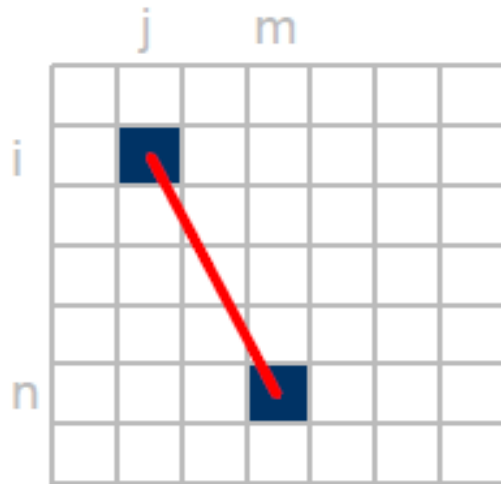
Cont. Example:

Case4: If $p_1 = 1$ and $p_3 = 1$

The length of the shortest m-path will be 4 (p, p_1, p_2, p_3, p_4)

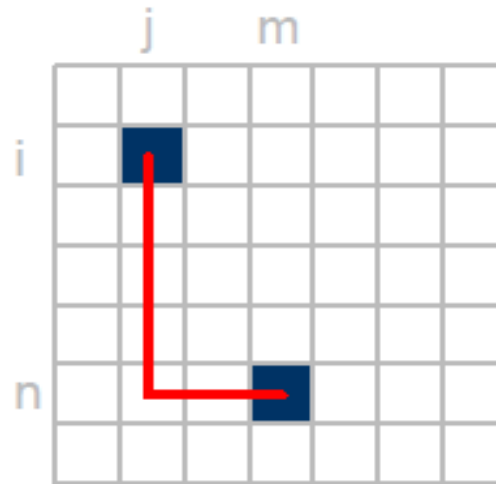
	1	1
1	1	
1		

Distance Measures



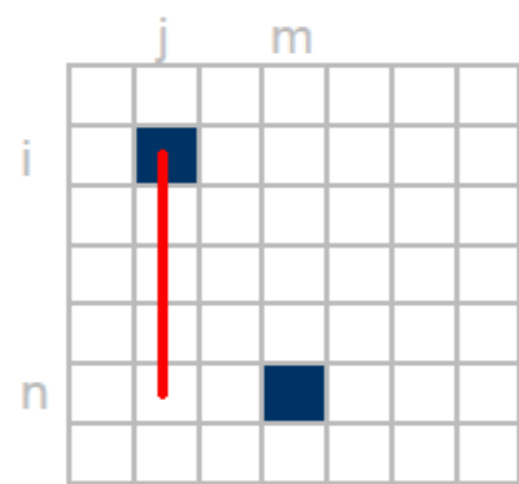
Euclidean Distance

$$= \sqrt{(i-n)^2 + (j-m)^2}$$



City Block Distance

$$= |i-n| + |j-m|$$



Chessboard Distance

$$= \max[|i-n|, |j-m|]$$

9. Linear and Nonlinear Operations

1. Linear operations
2. Non-linear operations

An operator is called a linear operator if it obeys the following rules of additivity and homogeneity. A non-linear operator, as the name suggests, does not follow these rules.

1. Property of additivity

$$\begin{aligned} H(a_1 f_1(x, y) + a_2 f_2(x, y)) &= H(a_1 f_1(x, y)) + H(a_2 f_2(x, y)) \\ &= a_1 H(f_1(x, y)) + a_2 H(f_2(x, y)) \end{aligned}$$

2. Property of homogeneity

$$= a_1 \times g_1(x, y) + a_2 \times g_2(x, y)$$

$$H(k f_1(x, y)) = k H(f_1(x, y)) = k g_1(x, y)$$

Image Vs Array Operations

Image operations are array operations. These operations are done on a pixel-by-pixel basis. Array operations are different from matrix operations. For example, consider two images

$$F_1 = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \text{ and } F_2 = \begin{pmatrix} E & F \\ G & H \end{pmatrix}$$

The multiplication of F_1 and F_2 is element-wise, as follows:

$$F_1 \times F_2 = \begin{pmatrix} AE & BF \\ CG & HD \end{pmatrix}$$

In addition, one can observe that $F_1 \times F_2 = F_2 \times F_1$, whereas matrix multiplication is clearly different, since in matrices, $A \times B \neq B \times A$. By default, image operations are array operations only.

Arithmetic operations - Addition

Two images can be added in a direct manner, as given by

$$g(x, y) = f_1(x, y) + f_2(x, y)$$

Table 3.1 Data type and allowed ranges

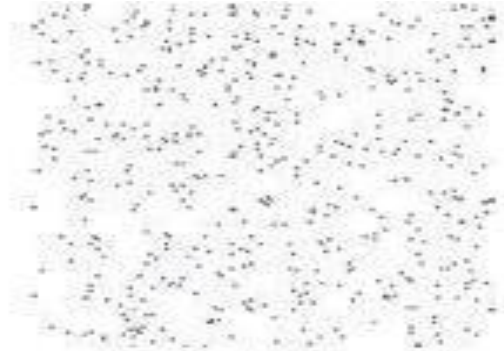
S. no.	Data type	Data range
1	Uint 8	0–255
2	Uint 16	0–65,535
3	Uint 32	0–4,29,49,67,295
4	Uint 64	0–1,84,46,74,40,73,70,95,51,615

Similarly, it is possible to add a constant value to a single image, as follows:

$$g(x, y) = f_1(x, y) + k$$



(a)



(b)



(c)



(d)

Fig. 3.14 Results of the image addition operation (a) Image 1 (b) Image 2
(c) Addition of images 1 and 2 (d) Addition of image 1 and constant 50

Image Subtraction

The subtraction of two images can be done as follows. Consider

$$g(x, y) = f_1(x, y) - f_2(x, y)$$

where $f_1(x, y)$ and $f_2(x, y)$ are two input images and $g(x, y)$ is the output image. To avoid negative values, it is desirable to find the modulus of the difference as

$$g(x, y) = |f_1(x, y) - f_2(x, y)|$$

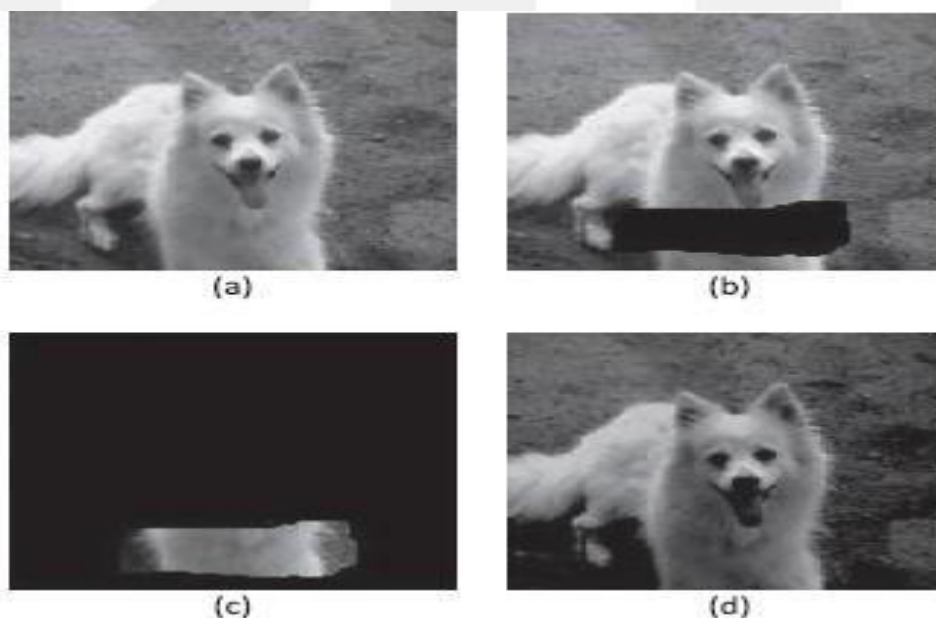


Fig. 3.15 Results of the image subtraction operation (a) Image 1 (b) Image 2 (c) Subtraction of images 1 and 2 (d) Subtraction of constant 50 from image 1

Image Multiplication

$$g(x, y) = f_1(x, y) \times f_2(x, y)$$

$$g(x, y) = f(x, y) \times k$$



Fig. 3.16 Result of multiplication operation (image $\times 1.25$) resulting in good contrast

Image Division

Similar to the other operations, division can be performed as

$$g(x, y) = \frac{f_1(x, y)}{f_2(x, y)}$$

where $f_1(x, y)$ and $f_2(x, y)$ are two input images and $g(x, y)$ is the output image.

$$g(x, y) = \frac{f(x, y)}{k}, \text{ where } k \text{ is a constant.}$$

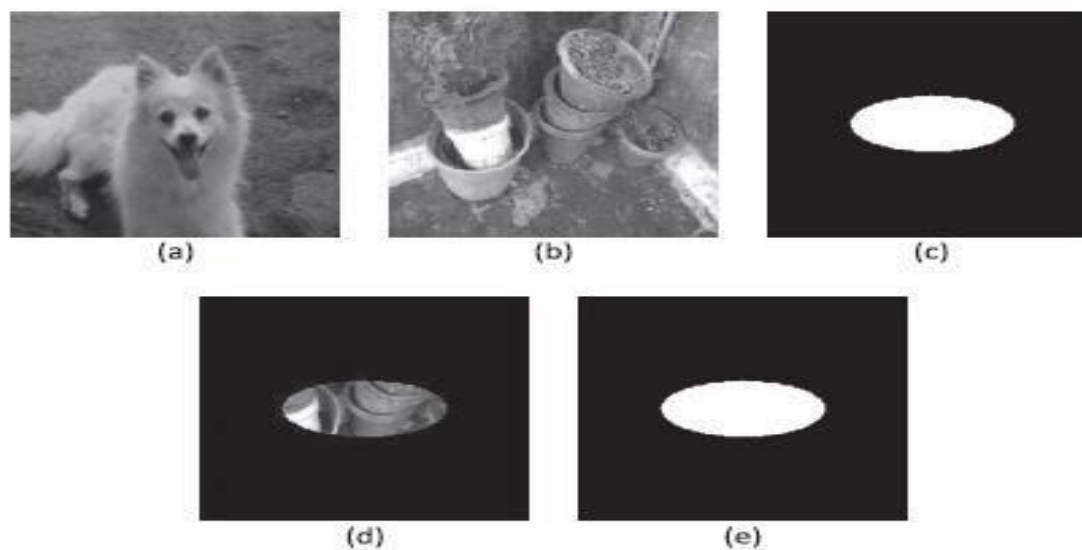


Fig. 3.17 Image division operation (a) Result of the image division operation (image/1.25)
 (b) Image 1 (c) Image 2 used as a mask (d) Image 3 = image 1 × image 2
 (e) Image 4 = image 3/image 1

References

1. Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, 3rd Edition Pearson Education Publisher.
2. Pratt William, *Digital Image Processing*, John Wiley & Sons
3. S Jayaraman, S Esakirajan and T Verakumar *Digital Image Processing*, McGraw Hill
4. John C. Russ, *The Image Processing Handbook*, by; CRC Press, 2007
5. A. K. Jain, *Fundamentals of Digital Image Processing*, Pearson Education
6. Mark Nixon and Alberto Aguado, *Feature Extraction and Image Processing*, Academic Press, 2008



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